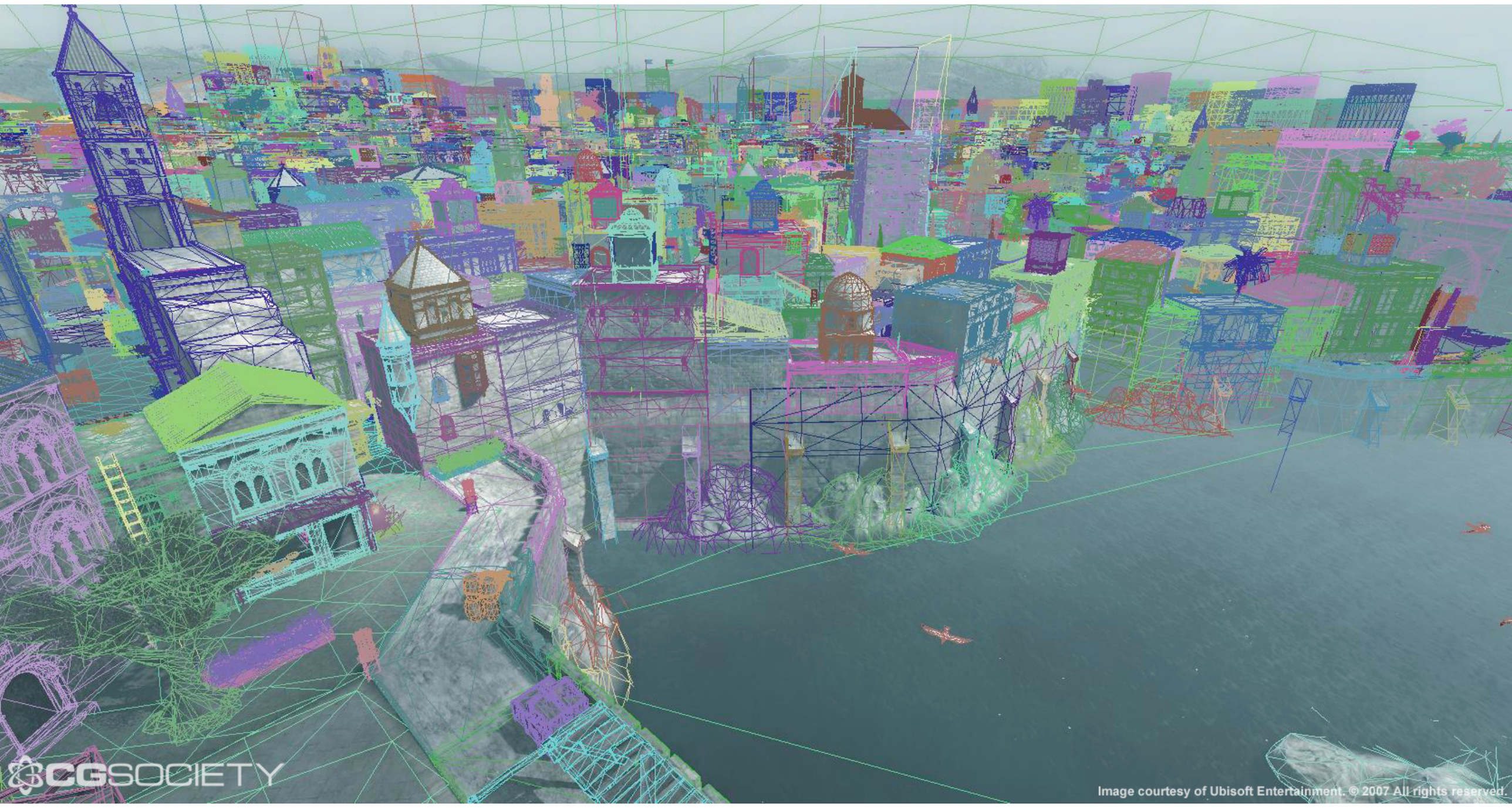
Representation of shape in graphics

Victor Bâcu

Representation of shape

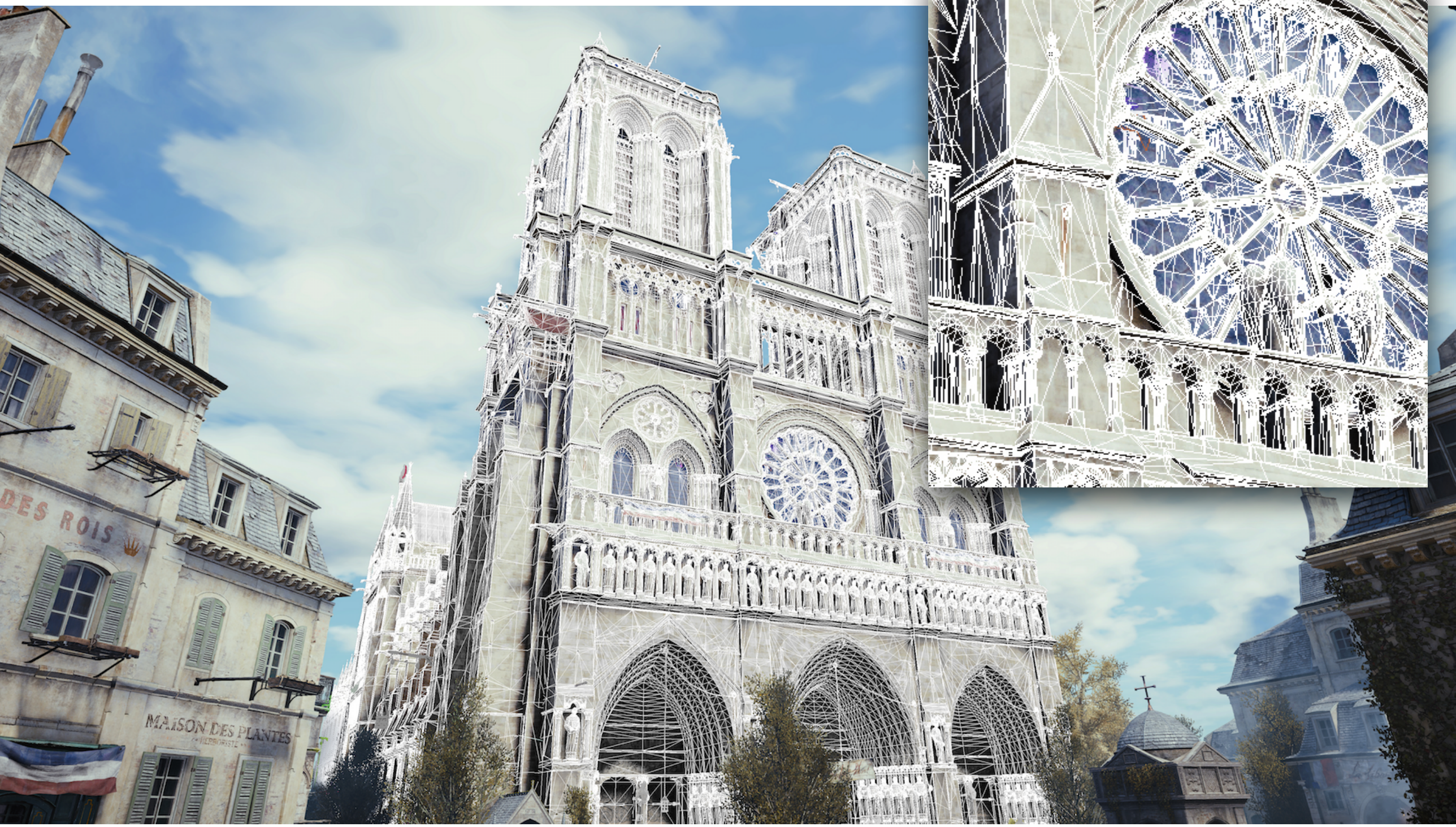
[http://www.cgsociety.org/stories/2007_10/assassins_creed/acre.jpg]

Representation of shape



[http://www.cgsociety.org/stories/2007_10/assassins_creed/acre-wire.jpg]

Representation of shape



[Assassin's Creed Unity]

3D object representations

Raw data	Surfaces	Solids	High-level structures
Point cloud	**Mesh**	**Voxel**	**Scene graph**
	Subdivision	**CSG**	**Skeleton**
	Parametric	**Sweep**	
	Implicit		

3D object representations

- **Triangle meshes** - the most used representation
- **Subdivision surfaces** - movies
- **Spline patches** - modeling
- **Constructive Solid Geometry (CSG)** - modeling machine parts
- **Volume data** - medical imaging
- **Point clouds** - computer vision

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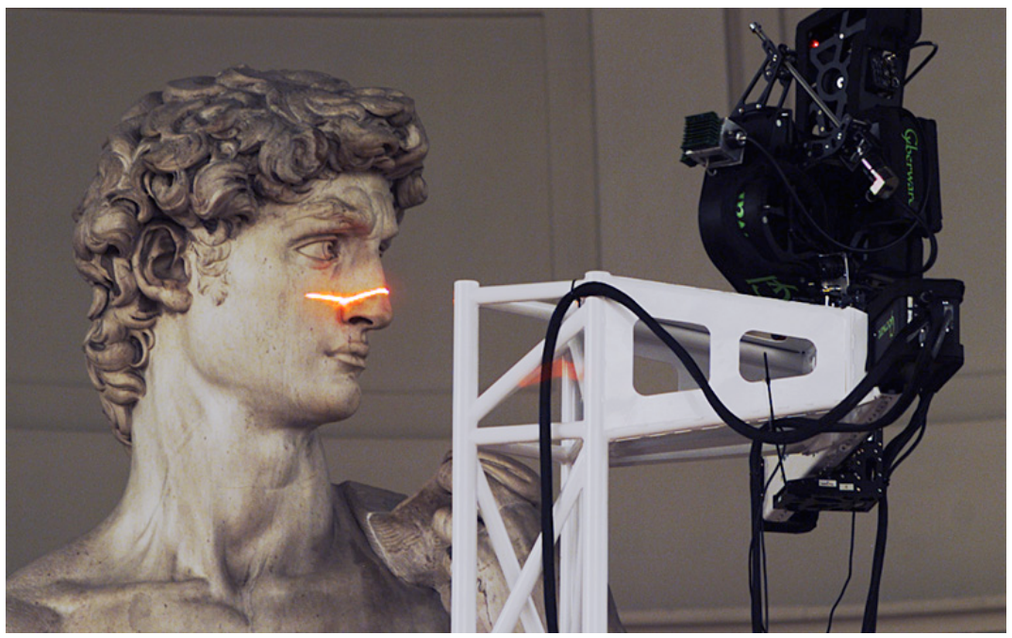
Point cloud

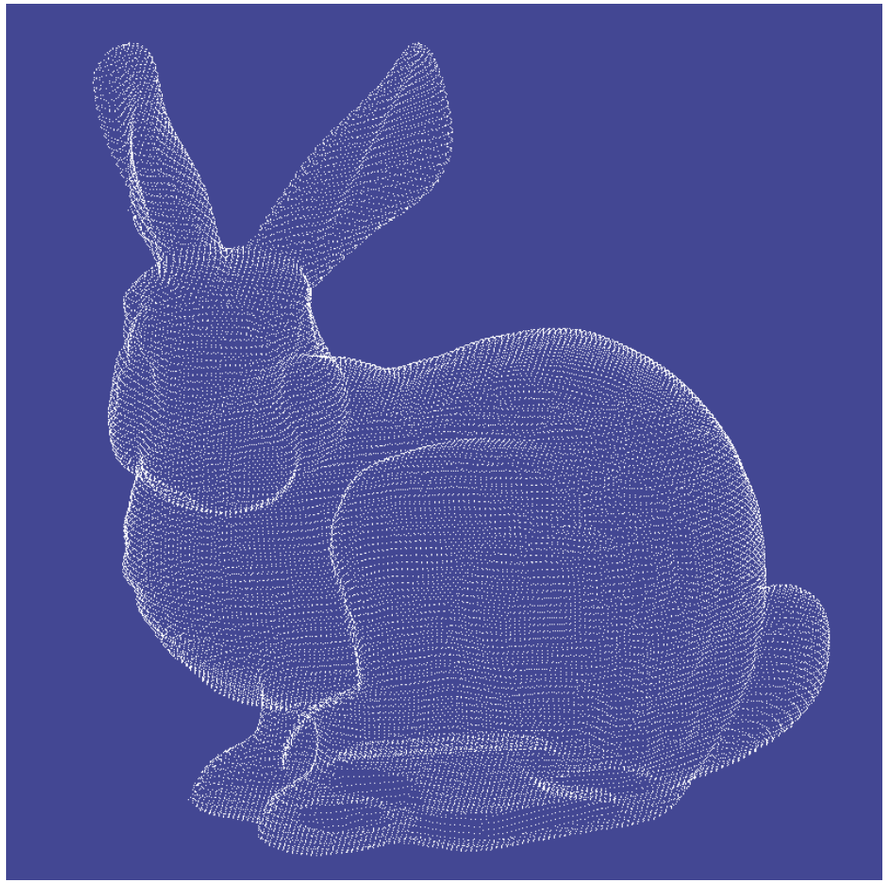
- Simplest representation, store only points without connectivity between them
- Collection of 3D coordinates (in some cases also the normals)
- Created using **3d scanning devices**



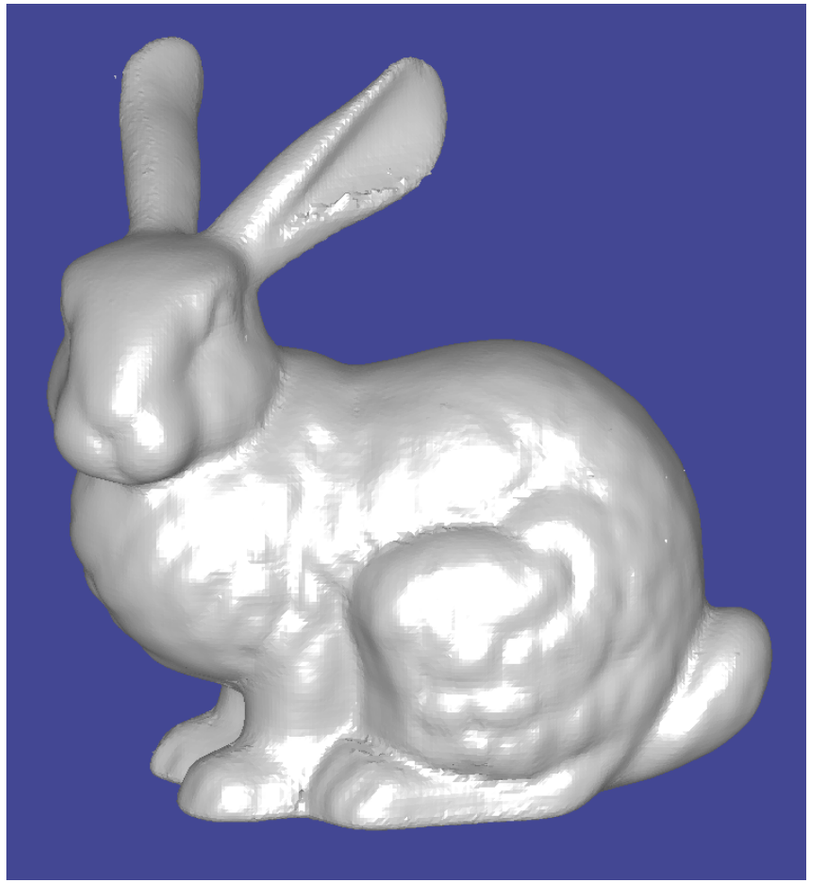
[Stanford, The Digital Michelangelo Project]

3D Point Cloud Reconstruction

- Issues: unstructured data
- Construct a polygonal (e.g. triangle mesh) representation of the point cloud
- 362,272 points (about 725,000 triangles) -> 35,947 vertices, 69,451 triangles



[http://www.ccs.neu.edu/home/gaob/images/bunny_pc.png]



[http://www.ccs.neu.edu/home/gaob/images/bunny_mc_s.png]

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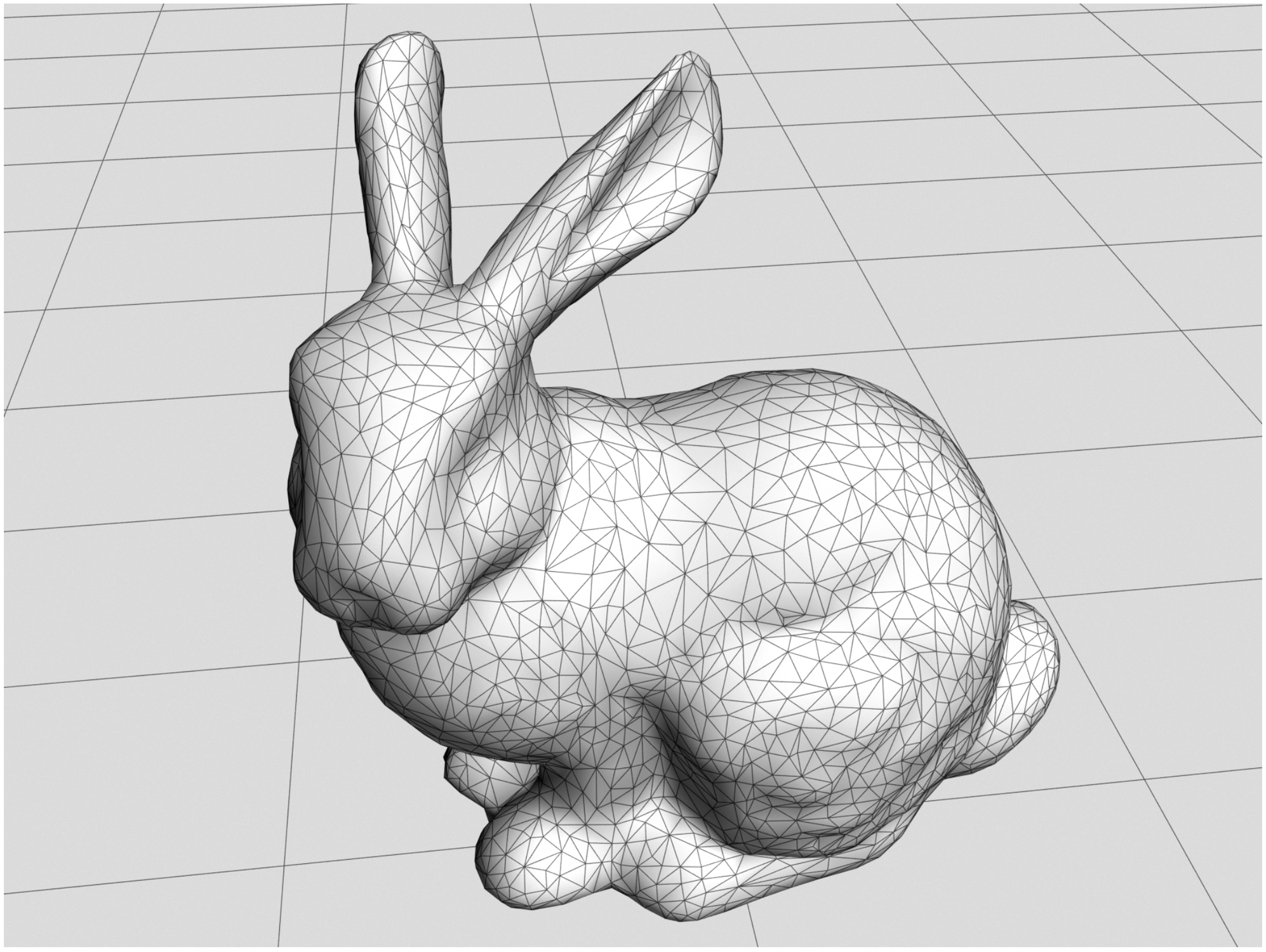
Polygonal mesh



Boundary representation

- Defined by:
  - **Geometric description**: specifies vertex coordinates, surface equations, etc.
  - **Topological description**: specifies connectivity of vertices, face orientation, etc.
- **Polygonal mesh**
  - Represent object surfaces using polygons
  - Uses different primitives: **vertices**, **edges**, and **faces**
  - Usually the 3D objects are represented by **triangle meshes**

Triangle mesh

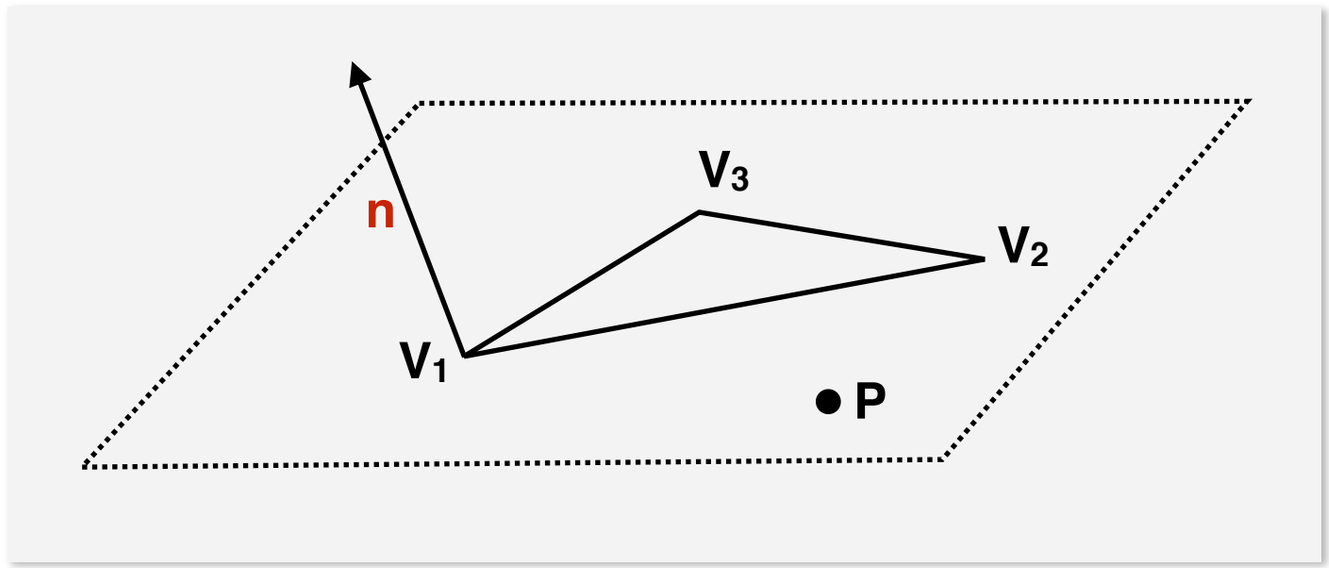
- Operations applied to triangle meshes:
  - **Subdivision**
    - replace a triangle by several smaller triangles
    - used to smooth out a mesh that has sharp points or edges
    - repeated subdivision operations increase the triangle count substantially; this can have a major impact on rendering performance
  - **Simplification**
    - a mesh is replaced by another mesh that's similar to it, topologically or geometrically

Triangles

- Always convex
- Fixed number of vertices
- The three vertices that define the triangle are on the same plane (no need to check planarity)
- Easy method to compute the normal vector (used to identify front/back faces)

$$\mathbf{n} = (V_2 - V_1) \times (V_3 - V_1)$$

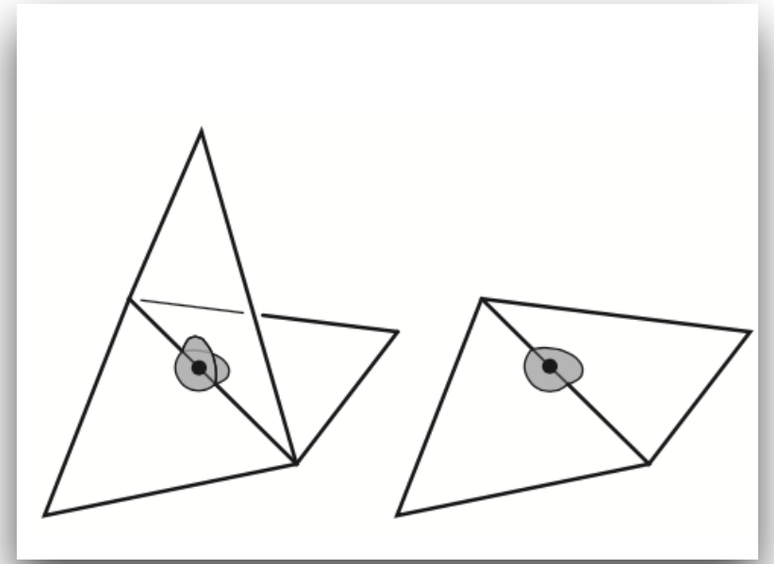
$$\mathbf{n} \cdot (P - V_1) = 0$$



Mesh topology

- The simplest and most restrictive requirement on the topology of a mesh is for the surface to be a manifold
- **Manifold mesh**
  - separates the space on the inside

    of the surface from the space outside
- Manifold **conditions**
  - Every edge is shared by exactly two triangles
  - Every vertex has a single, complete loop of triangles around it
- Manifold with **boundary conditions**
  - Every edge is used by either one or two triangles
  - Every vertex should have one connected fan of triangles

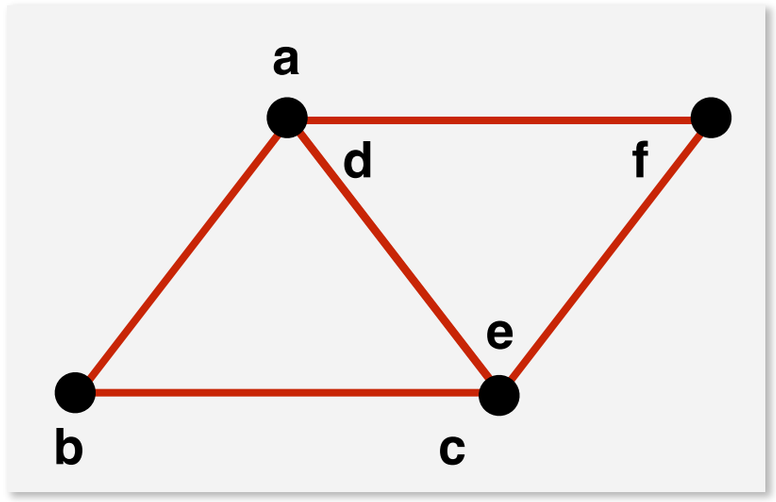


Data structures

Polygon-soup

- consist of a **list of polygons** (triangles), and each polygon contains the vertices coordinates
- very simple to render (process each individual polygon)
- **issues**:
  - multiple specifications of a particular vertex
  - no info about which triangle uses a particular vertex
  - no info about adjacent triangles

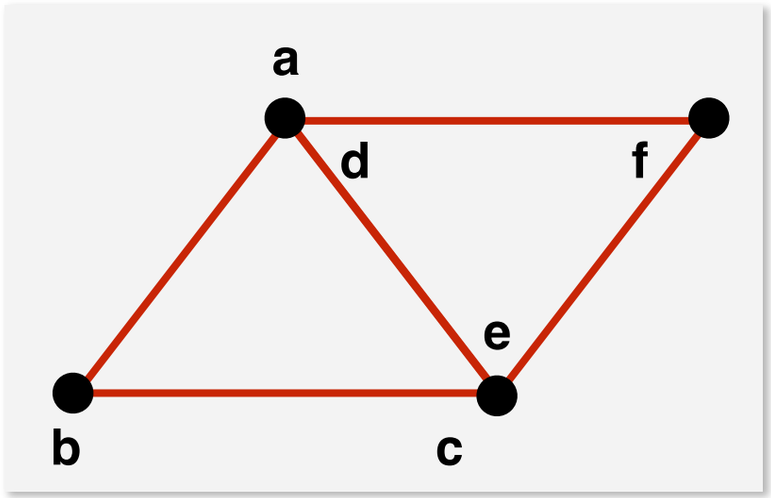
triangle #	vertex 1	vertex 2	vertex 3
0	(a_x, a_y, a_z)	(b_x, b_y, b_z)	(c_x, c_y, c_z)
1	(d_x, d_y, d_z)	(e_x, e_y, e_z)	(f_x, f_y, f_z)



Data structures

Indexed face set

- uses a list of vertices and a list of polygons
- each vertex is defined only once
- **issues**:
  - find all triangles that share a common vertex
  - find all triangles for all vertices
- store more connectivity information
  - triangles sharing a given vertex
  - triangles which are adjacent (sharing an edge)
  - vertices adjacent to a given vertex

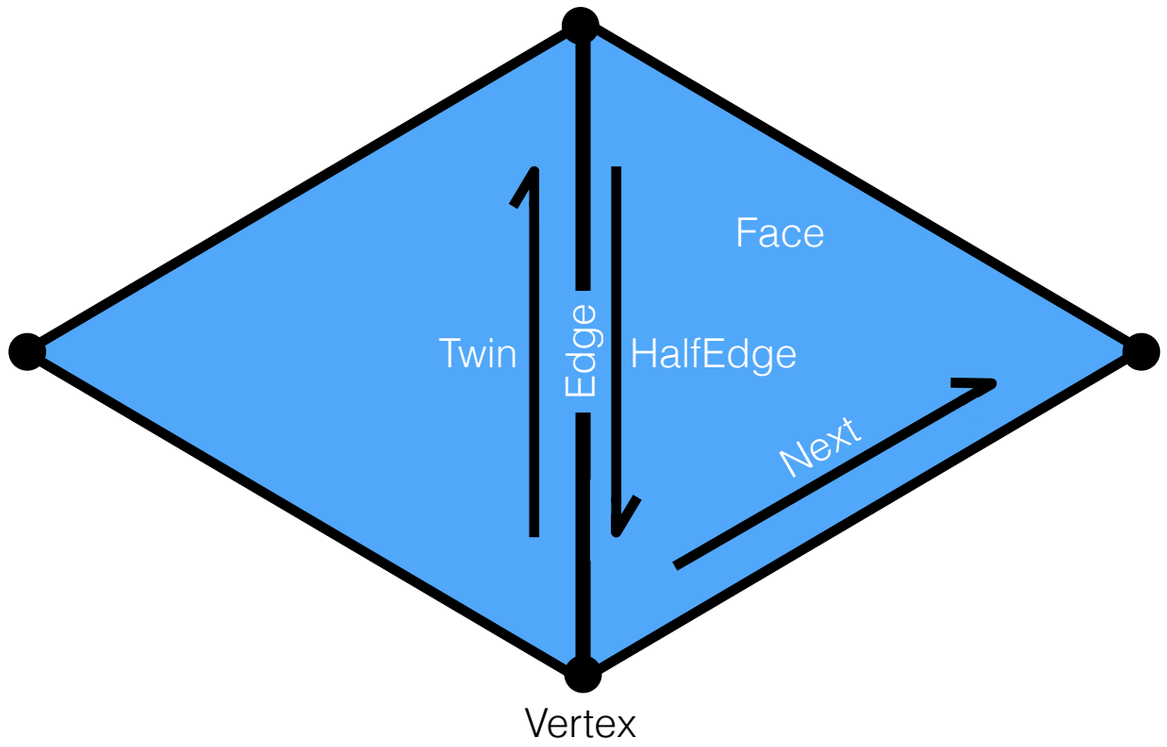


vertex #	x	y	z
0	a_x	a_y	a_z
1	b_x	b_y	b_z
2	c_x	c_y	c_z
3	f_x	f_y	f_z

triangle #	v_1	v_2	v_3
0	0	1	2
1	0	2	3

Half-edge data structure

```
struct HalfEdge
{
    HalfEdge* twin;
    HalfEdge* next;
    Vertex* vertex;
    Edge* edge;
    Face* face;
};
```




```
struct Egde
{
    HalfEdge*
    halfedge;
    …
};
```

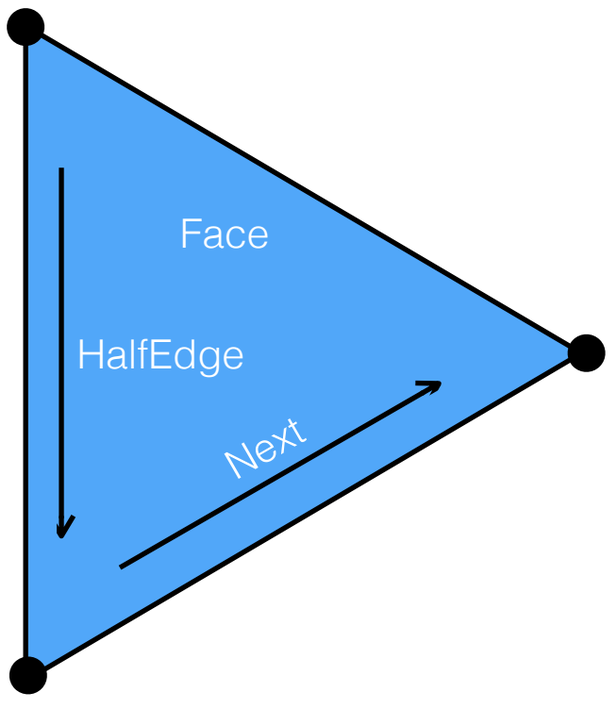
```
struct Face
{
    HalfEdge* halfedge;
    …
};
```

```
struct Vertex
{
    HalfEdge* halfedge;
    …
};
```

Mesh traversal

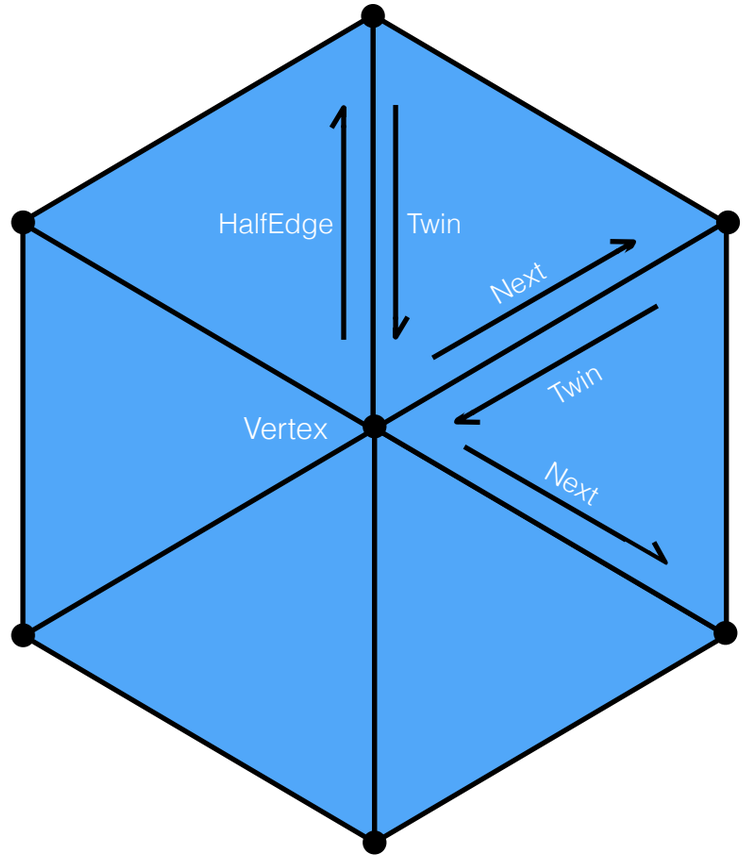
Visit all vertices of a face

```
HalfEdge* h = f->halfedge;
do{
    h = h->next;
    //process h->vertex
}while( h != f->halfedge );
```

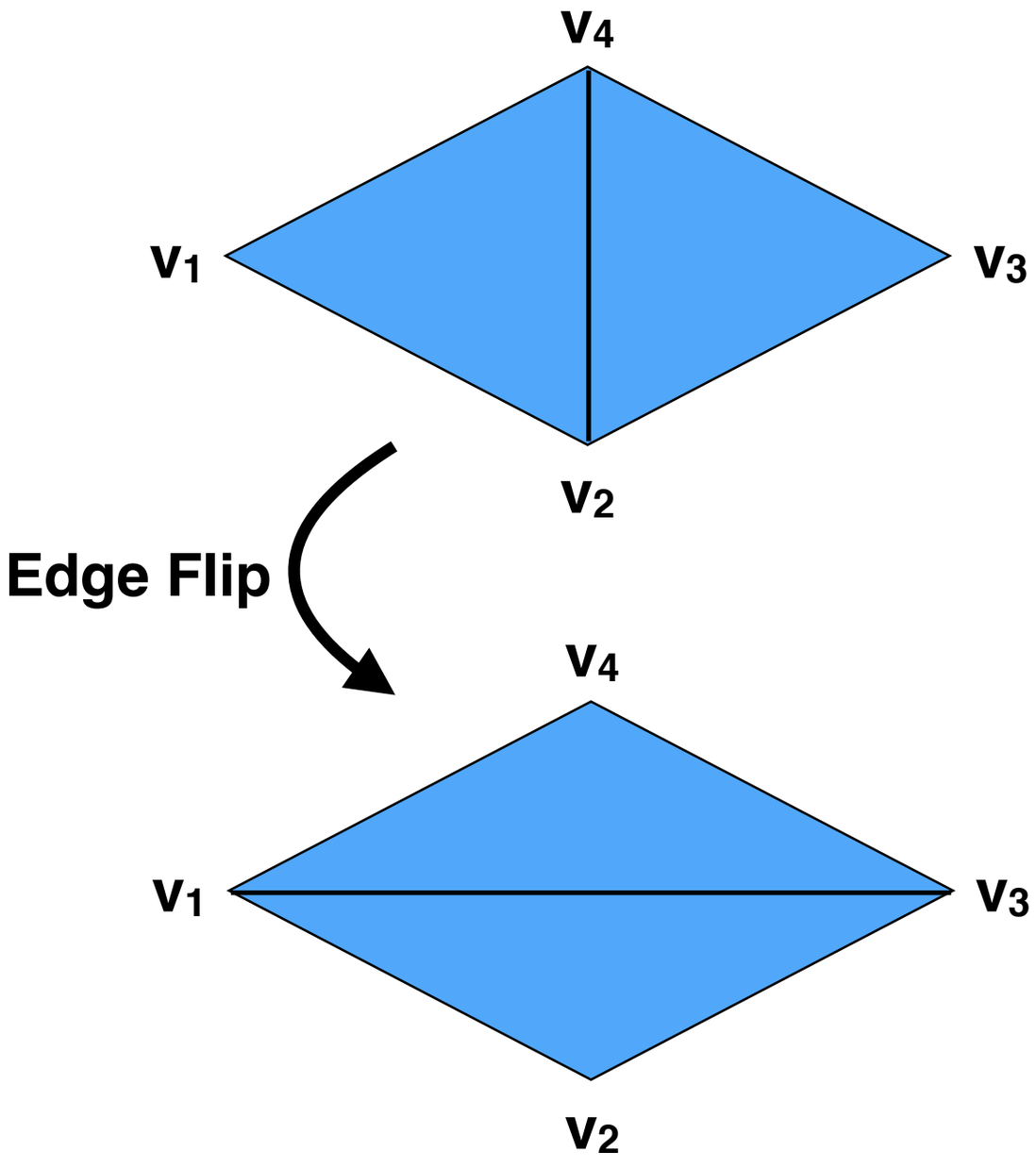


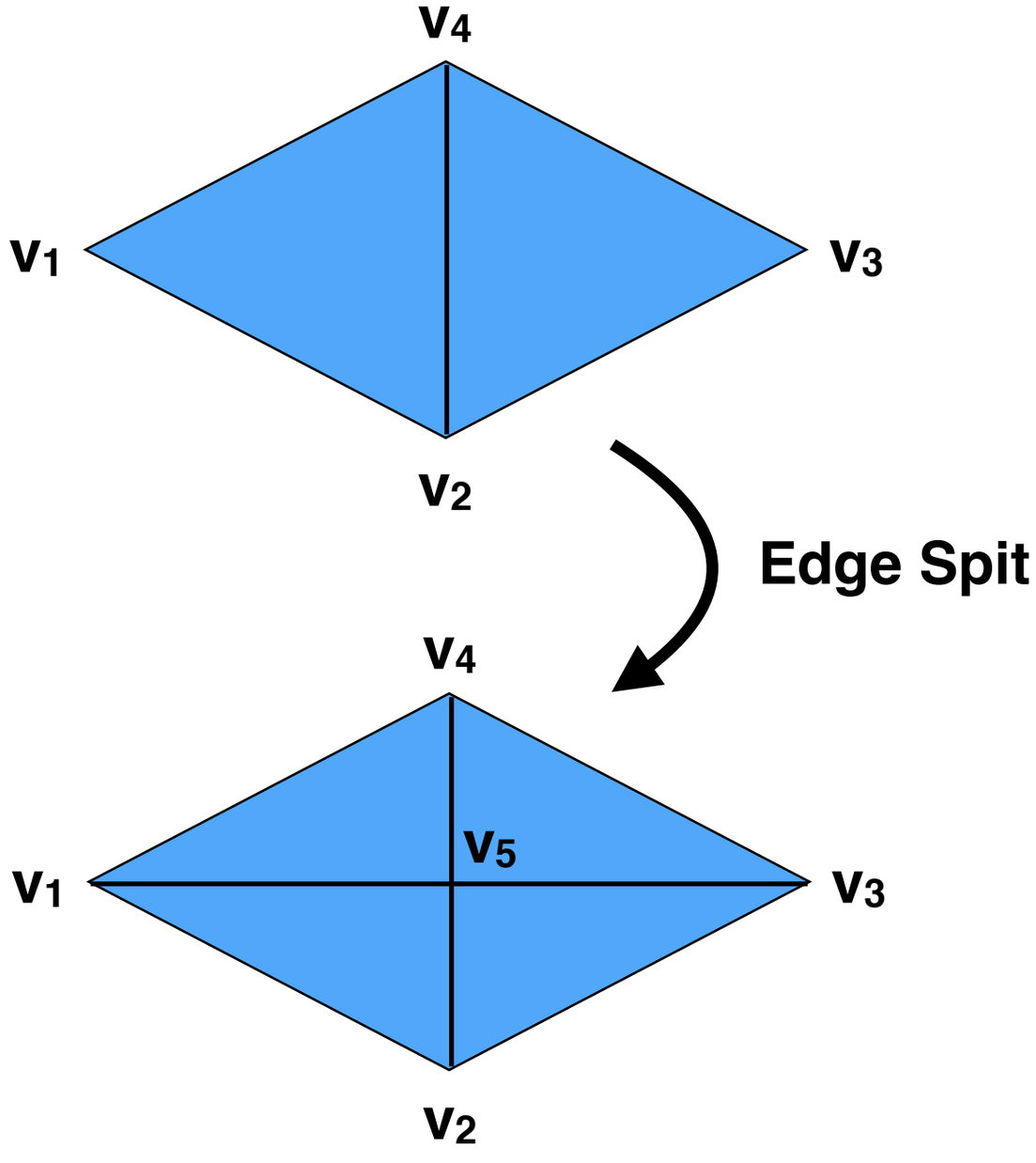
Visit all neighbors of a vertex

```
HalfEdge* h = v->halfedge;
do{
    h = h->twin->next;
    //process h->vertex
}while( h != v->halfedge );
```



Operations on HalfEdges





3D object representations

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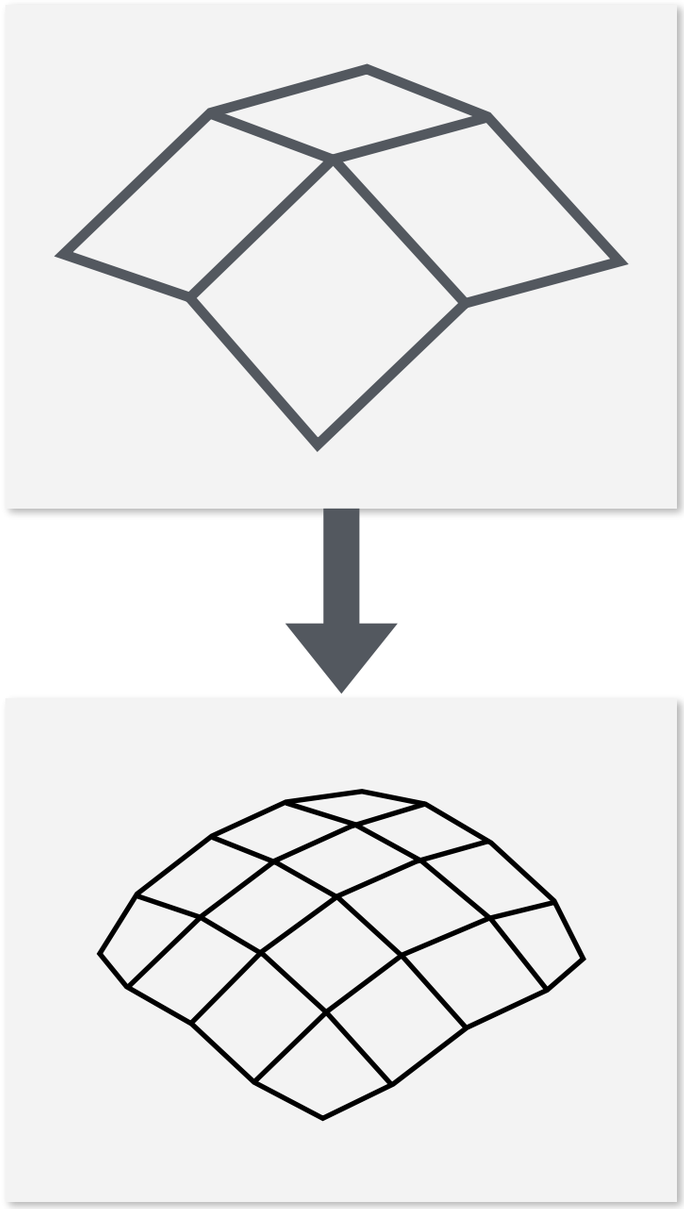
	Parametric	**Sweep**	
	Implicit		

Subdivision surfaces

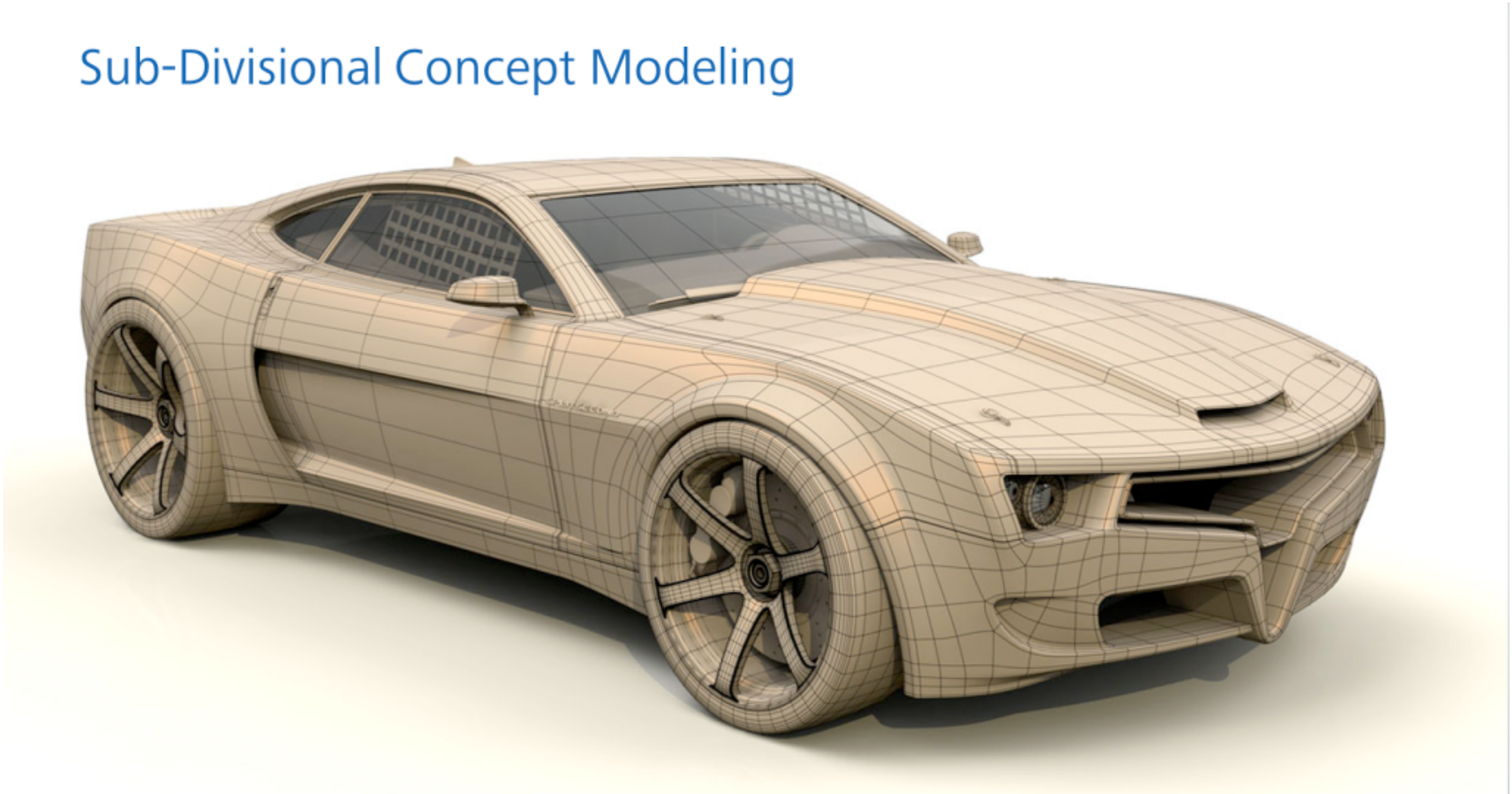
- A method for representing smooth surfaces by specifying a coarse polygonal mesh
- Subdivision surfaces are defined recursively
- Each step of refinement (iteration) adds new faces and vertices



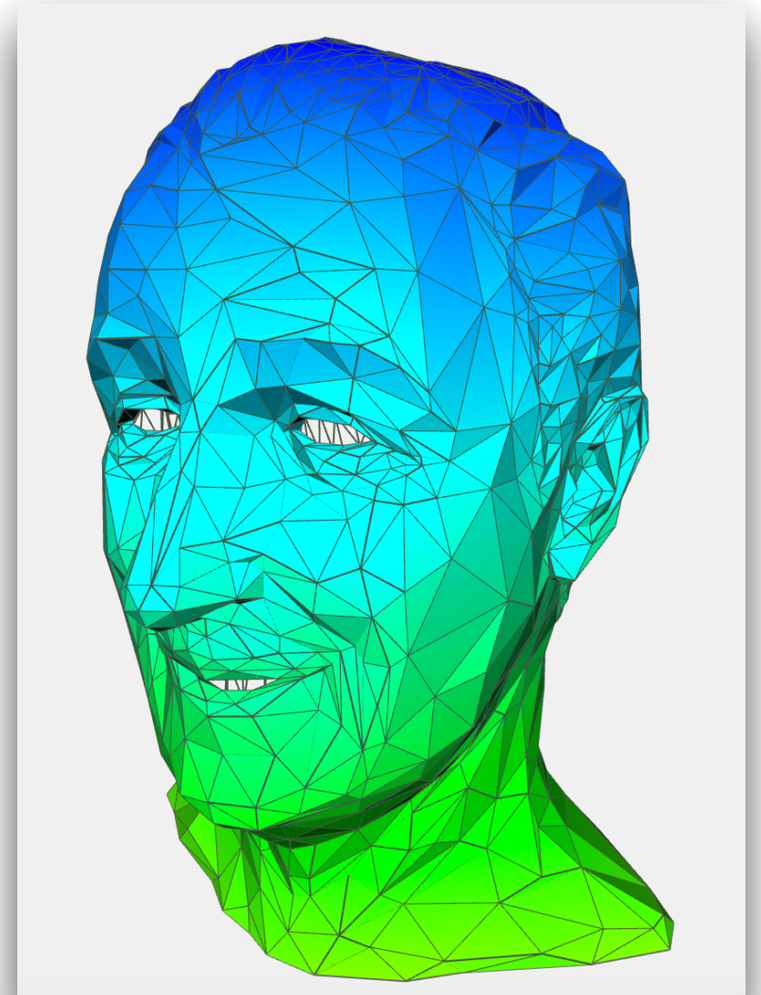
- The surfaces converges to a smooth limit surface (the “mathematical” result after infinite refinement steps)
- Subdivision schemes

	Triangles	Quads
Approximating	Loop	Catmull-Clark
Interpolating	Butterfly	Kobbelt

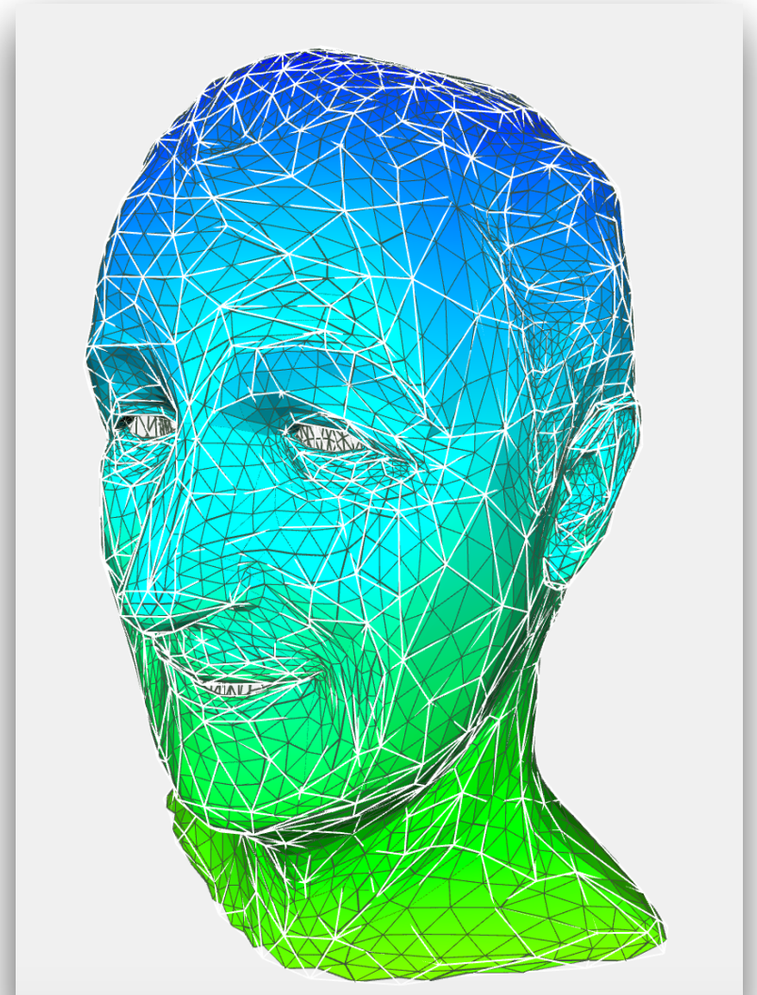
Subdivision surfaces



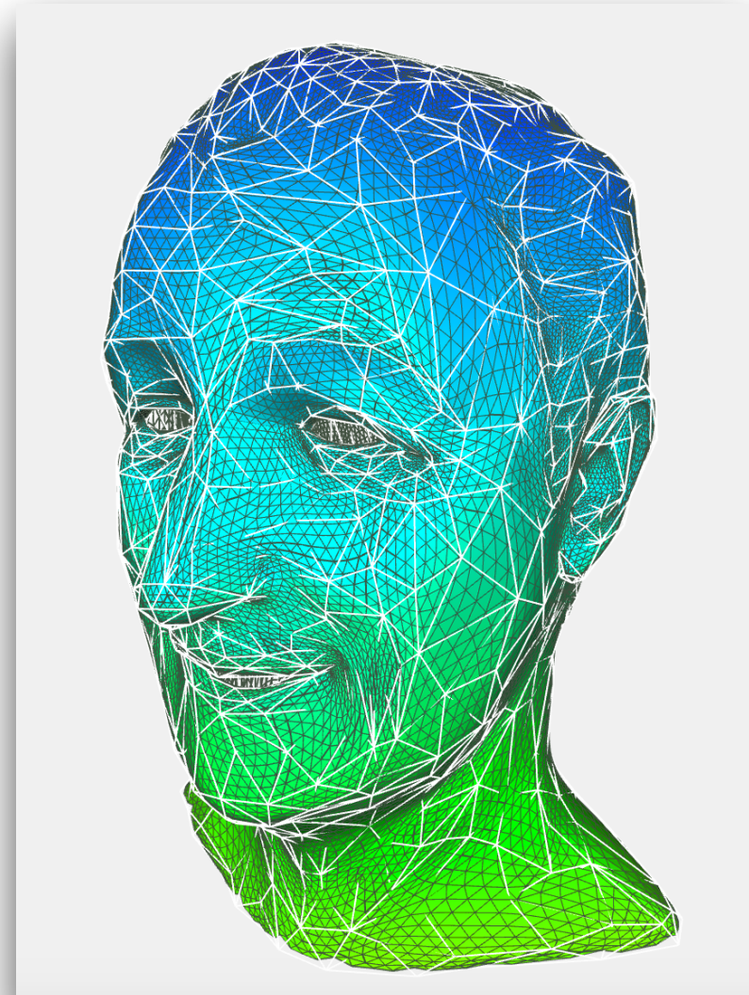
Subdivision surfaces



Vertices count: **1642**
Face count: **3232**



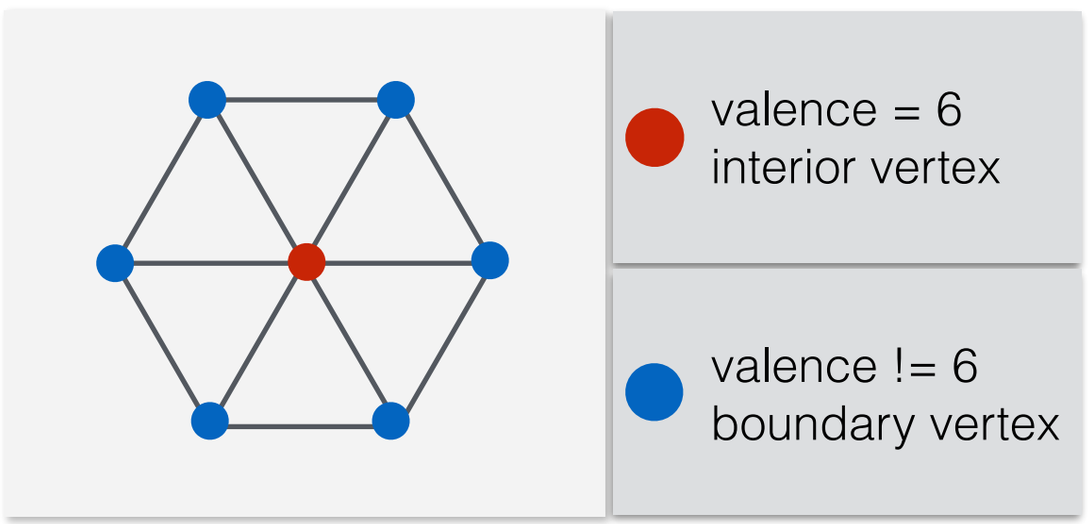
Vertices count: **6524**
Face count: **12928**

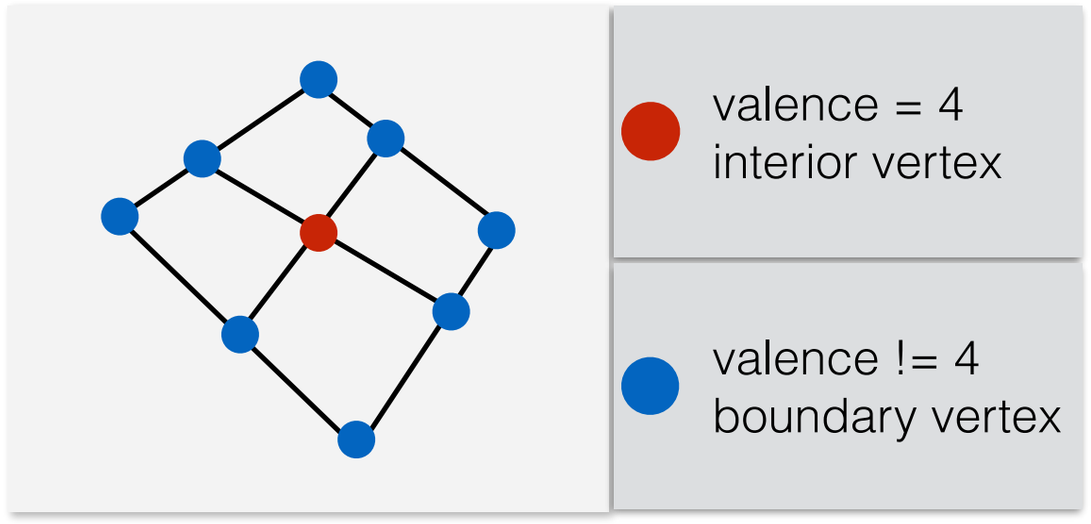


Vertices count: **25984**
Face count: **51712**

Subdivision concepts

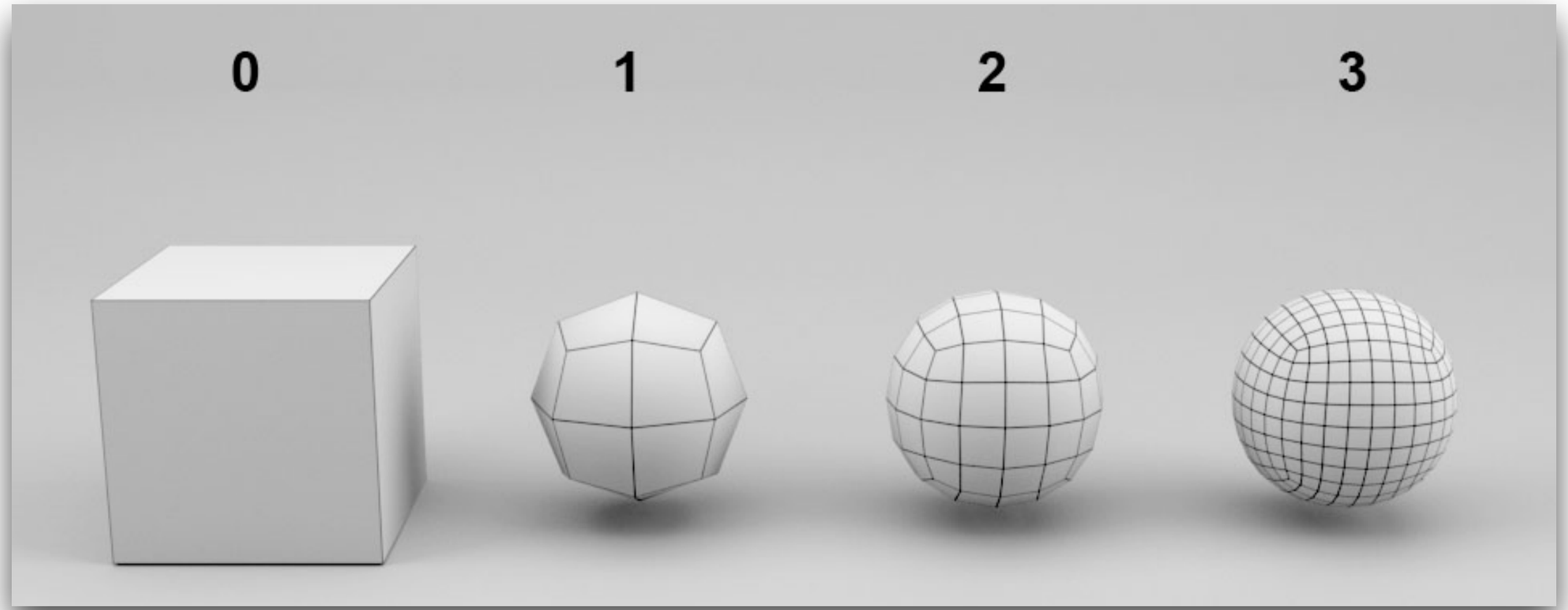
- **Valence/degree** of a vertex
  - the number of edges connected to it
- **Extraordinary** vertex
  - vertex which has a valence different from ordinary vertices
  - by applying subdivision the number of extraordinary points does not change





Catmull-Clark subdivision

- Works for arbitrary polygonal mesh
- The new mesh (after the first iteration) will consist only of **quadrilaterals**
- **Smoothness** of limit surface:
  - C^1 - at extraordinary vertices
  - C^2 - at ordinary vertices

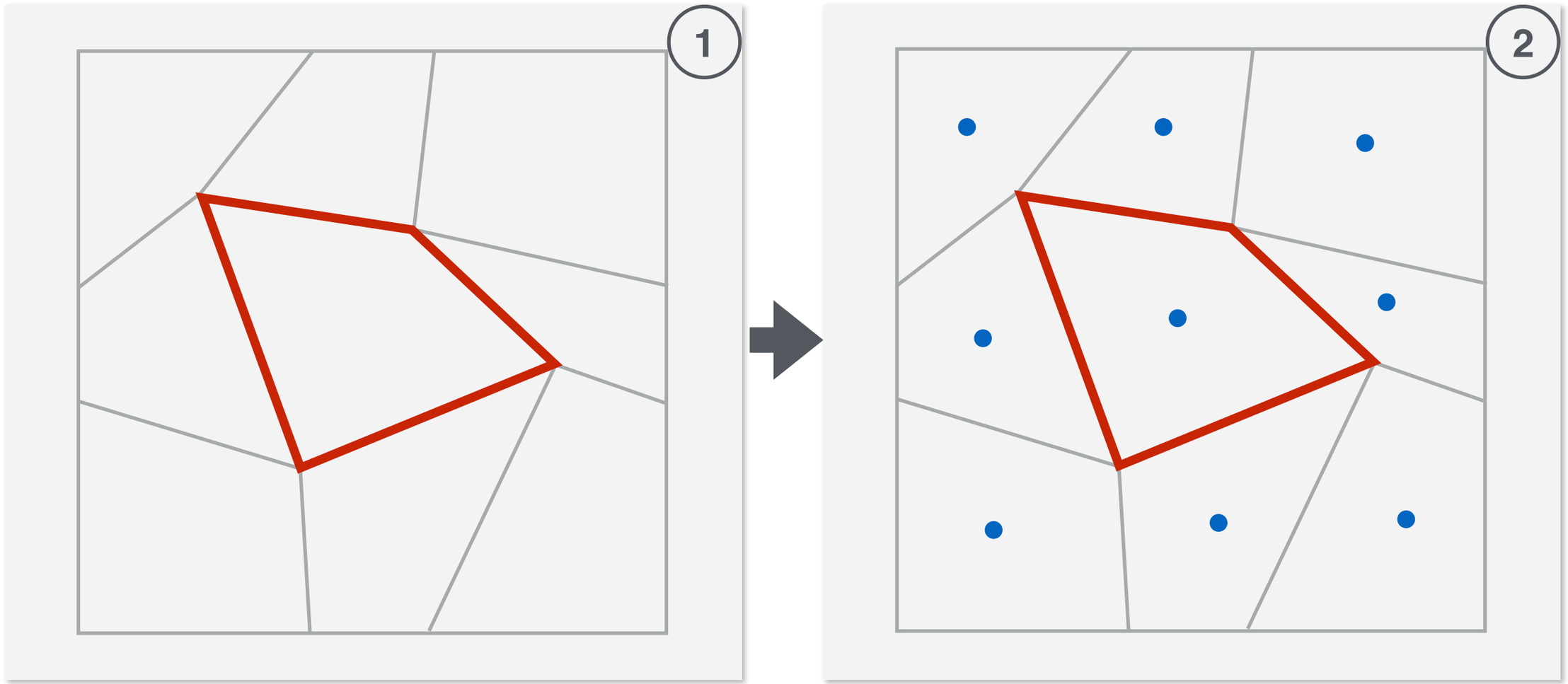


[https://support.solidangle.com/download/attachments/1081443/subdiv_IterationsCatclark.png?api=v2]

Catmull-Clark subdivision

Add new face points

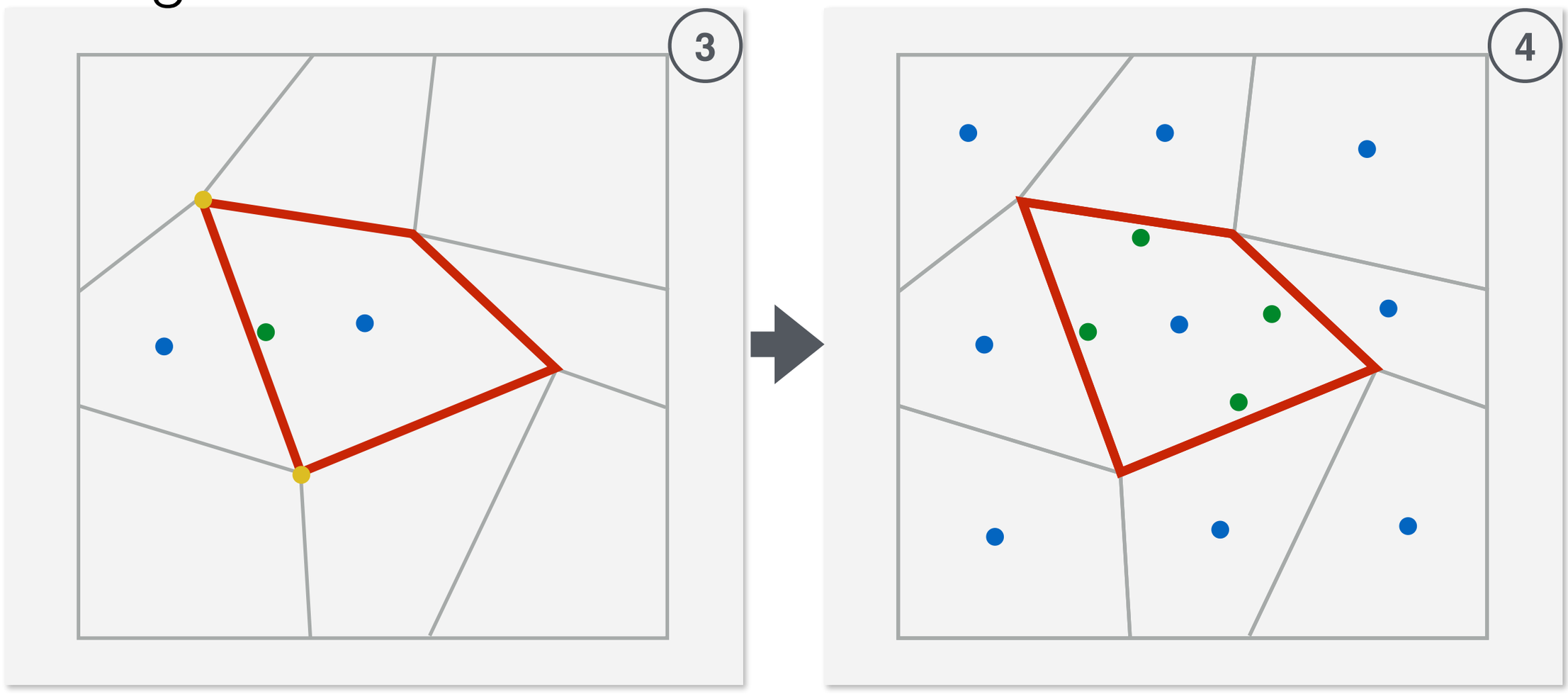
- the average of the points defining the face



Catmull-Clark subdivision

Add new edge points

- the average of the new face points of the faces sharing the edge and the original endpoints of the edge

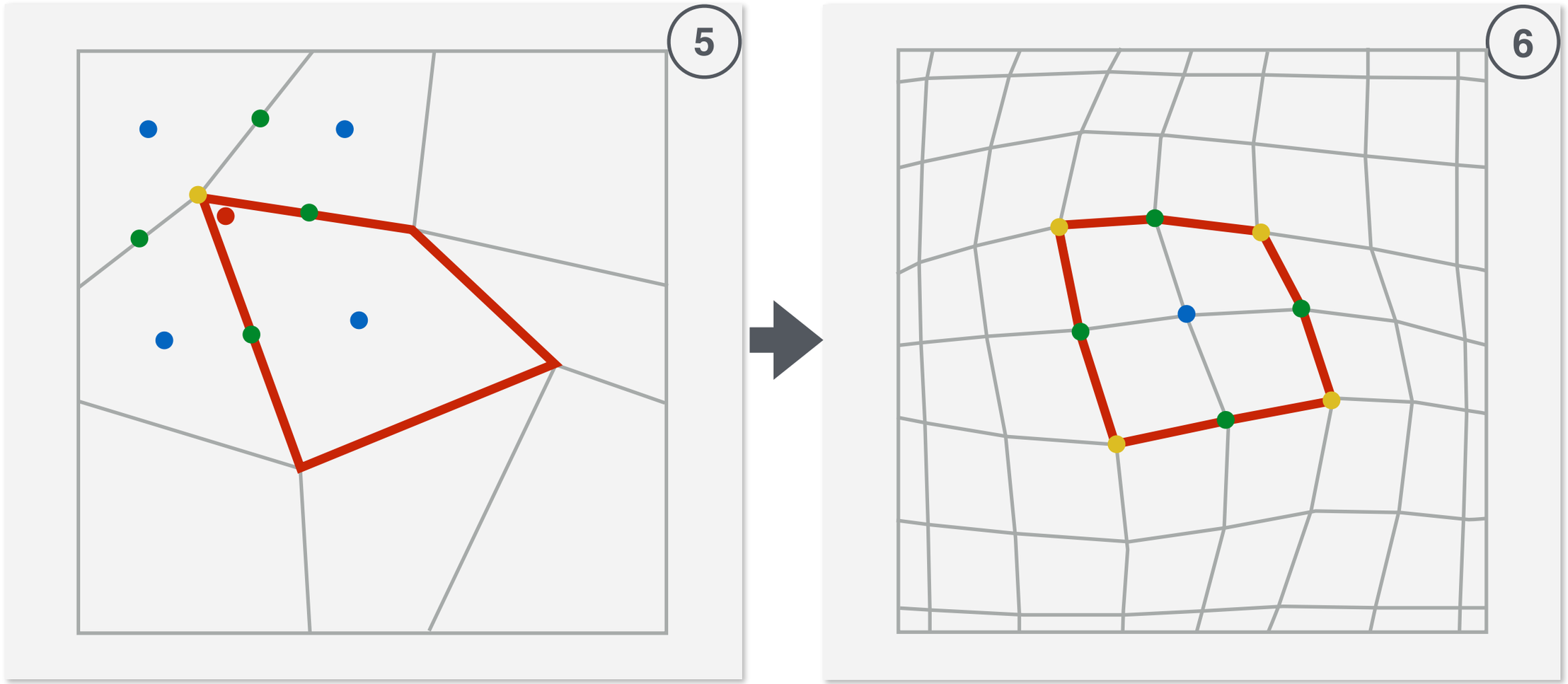


Catmull-Clark subdivision

Move the original point P

$$\frac{F + 2R + (n-3)P}{n}$$

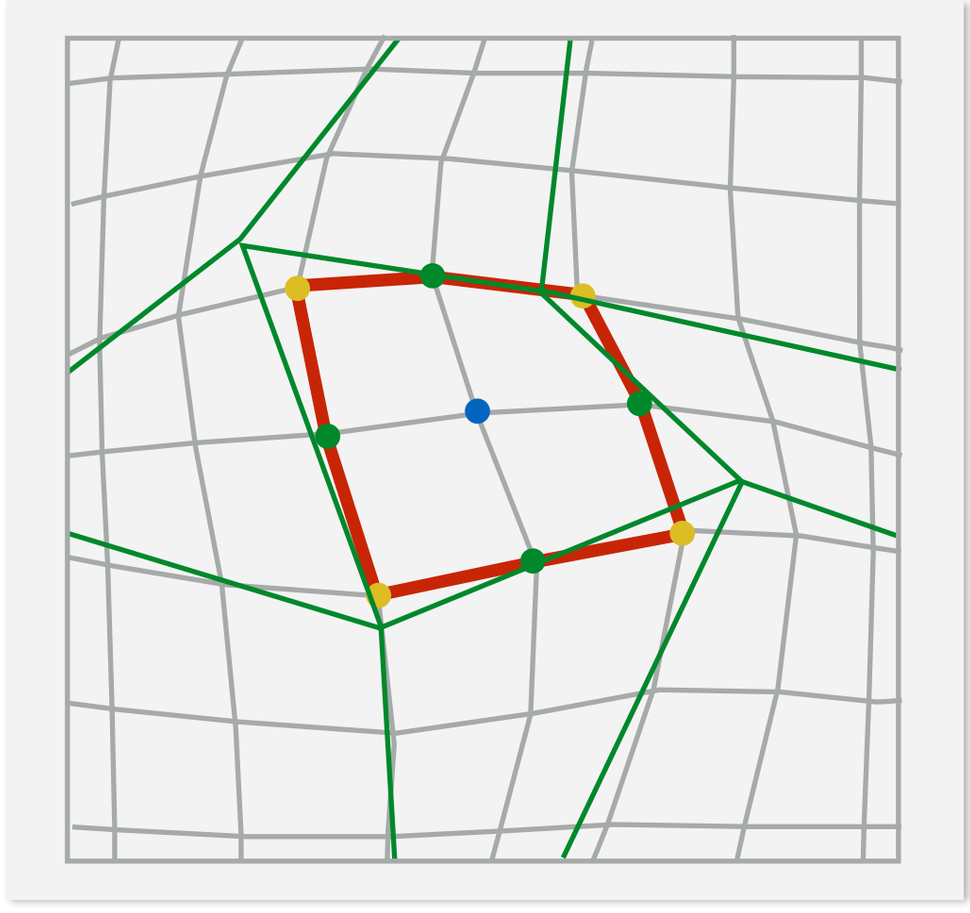
- **F** - the average of all **n** new face points (faces containing point P)
- **R** - the average of all **n** edge midpoints (edges containing point P)
- **P** - the original point



Catmull-Clark subdivision

Modify the topology of the original model

- For each new face point, add new edges connecting the new face point to each new edge point of the face
- Connect each new vertex point to the new edge points
- Define new faces



Catmull-Clark subdivision

- **Add new face points** - the average of the points defining the face
- **Add new edge points** - the average of the new face points of the faces sharing the edge and the original endpoints of the edge
- **Move the original point P**

$$\frac{F + 2R + (n-3)P}{n}$$

  - **F** - the average of all **n** new face points (faces containing point P)
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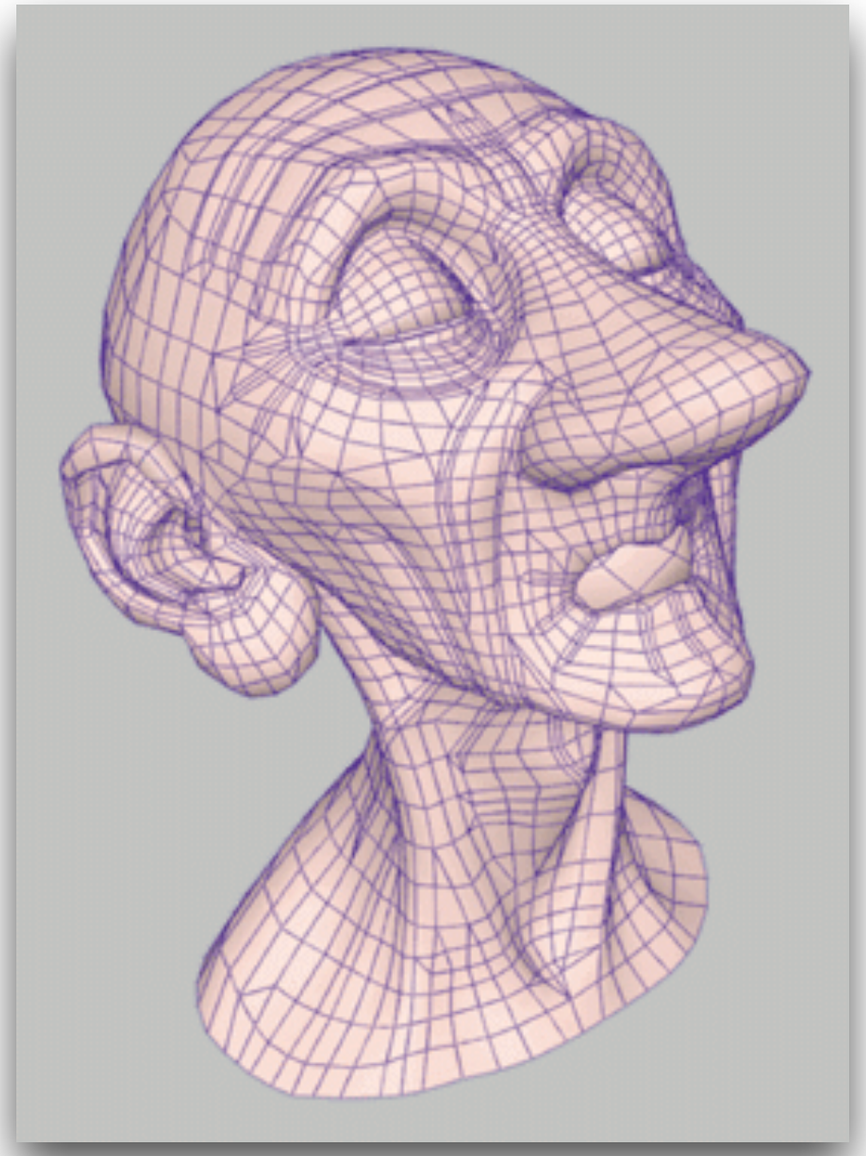
  - Define new faces

Catmull-Clark example

- Pixar - Geri's Game



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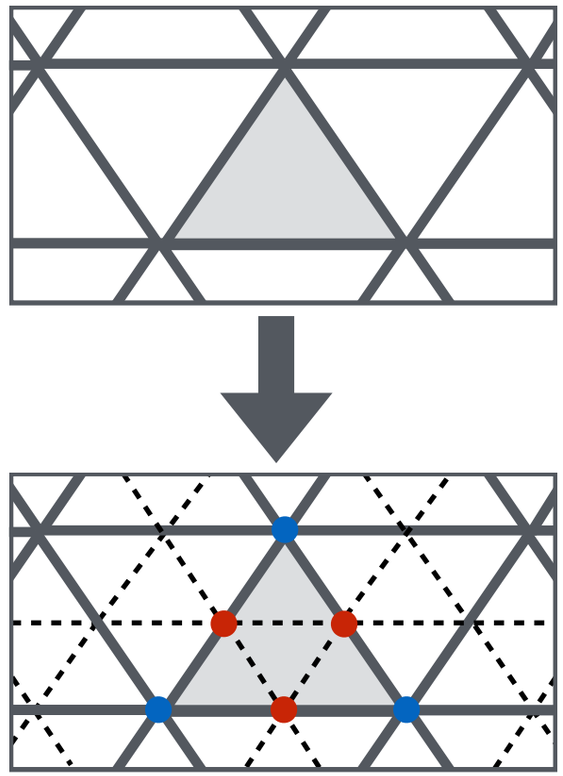
Loop subdivision

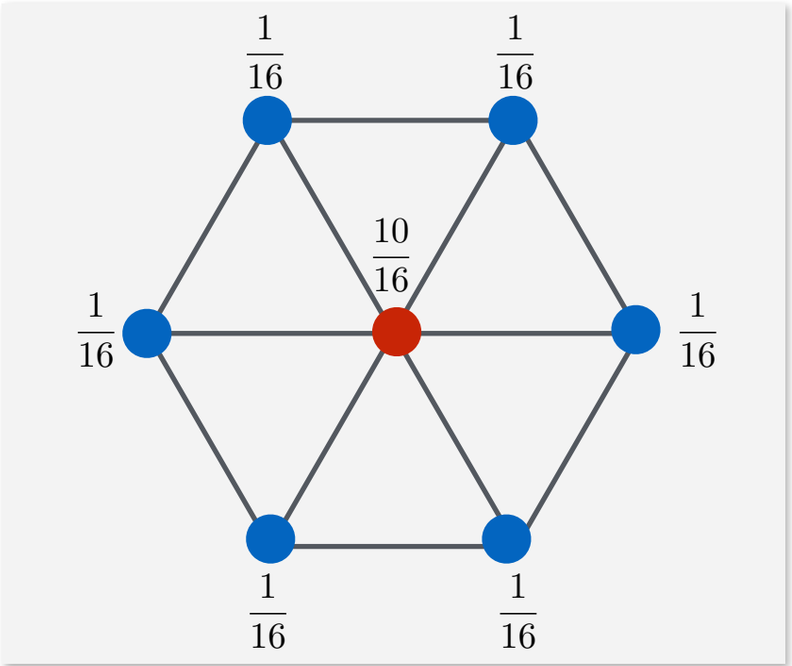
- Works on triangles
- Divides each triangle into four smaller ones
- **Smoothness** of limit surface:
  - C^1 - at extraordinary vertices
  - C^2 - at ordinary vertices



- Loop subdivision scheme:
  - **Refinement** - subdivide each triangle into 4 new triangles by splitting each edge and connecting new vertices
  - **Smoothing** - adjust the locations of the existing vertices and of the new inserted vertices

Loop subdivision

- Move each of the original vertices to new location - weighted average of the original vertex and its neighbours vertices

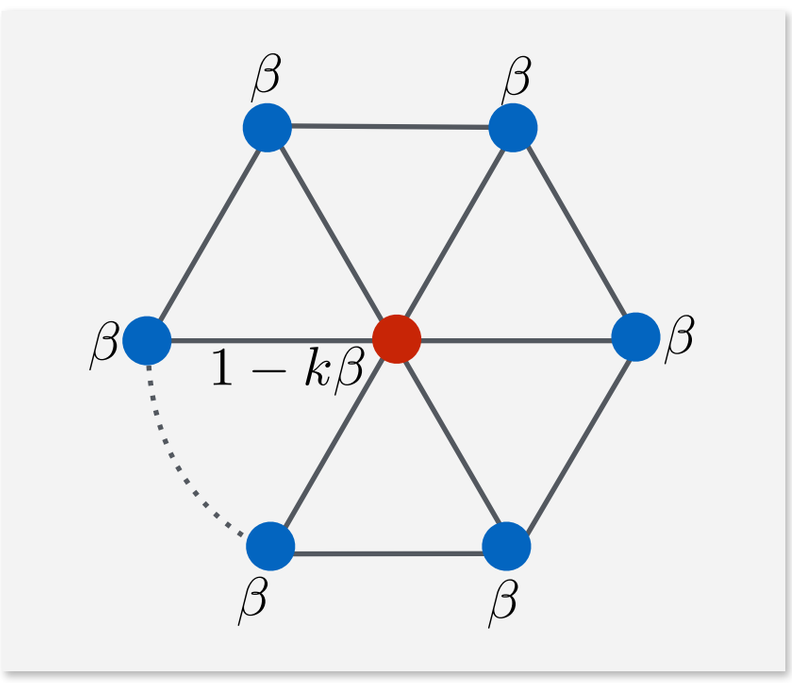


$$V_{new} = (1 - k\beta) * V_{old} + \sum(\beta * OriginalAdjacentVertex)$$

- Original loop: $\beta = \frac{1}{k}(\frac{5}{8} - (\frac{3}{8} + \frac{1}{4}\cos\frac{2\pi}{k})^2)$

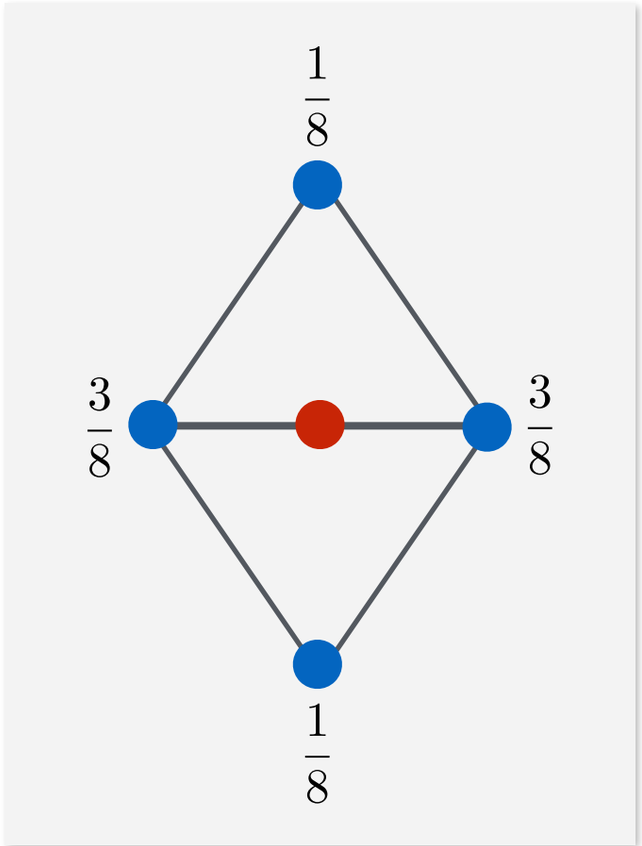
- Warren: $\beta = \frac{3}{8k}, n > 3$

  $\beta = \frac{3}{16}, n = 3$



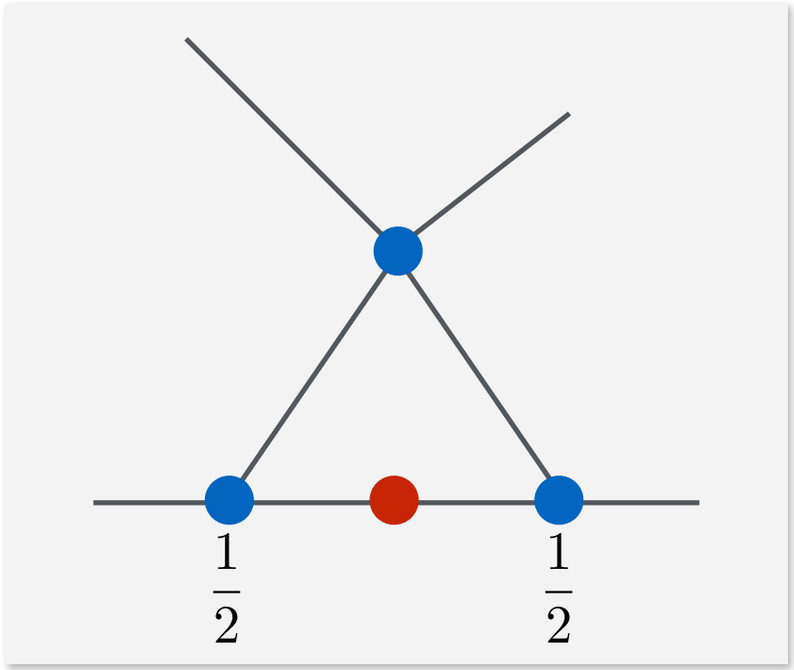
Loop subdivision

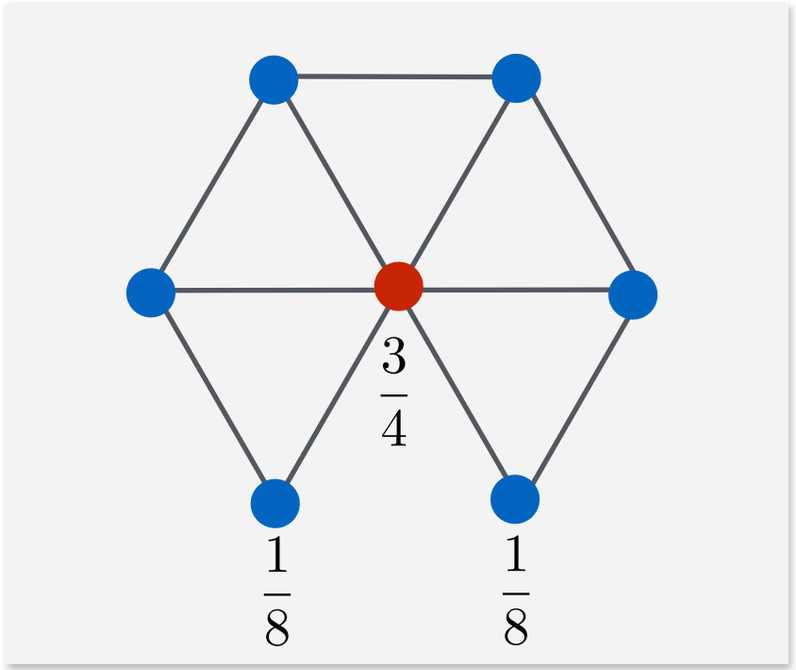
- Choose the location for the new vertices as a weighted average of the original vertices in local neighbourhood



Loop subdivision

- Rules for crease and boundary:





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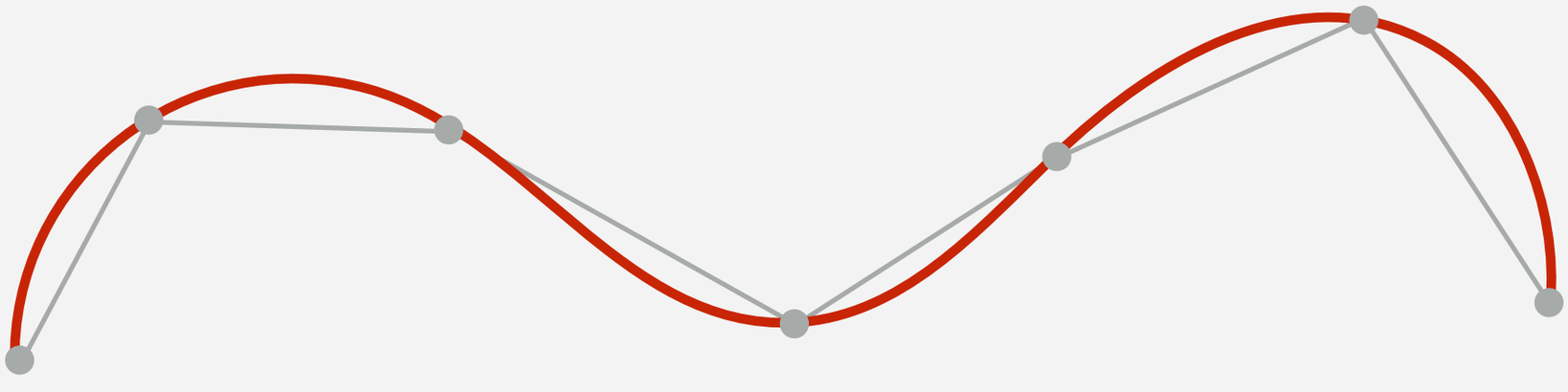
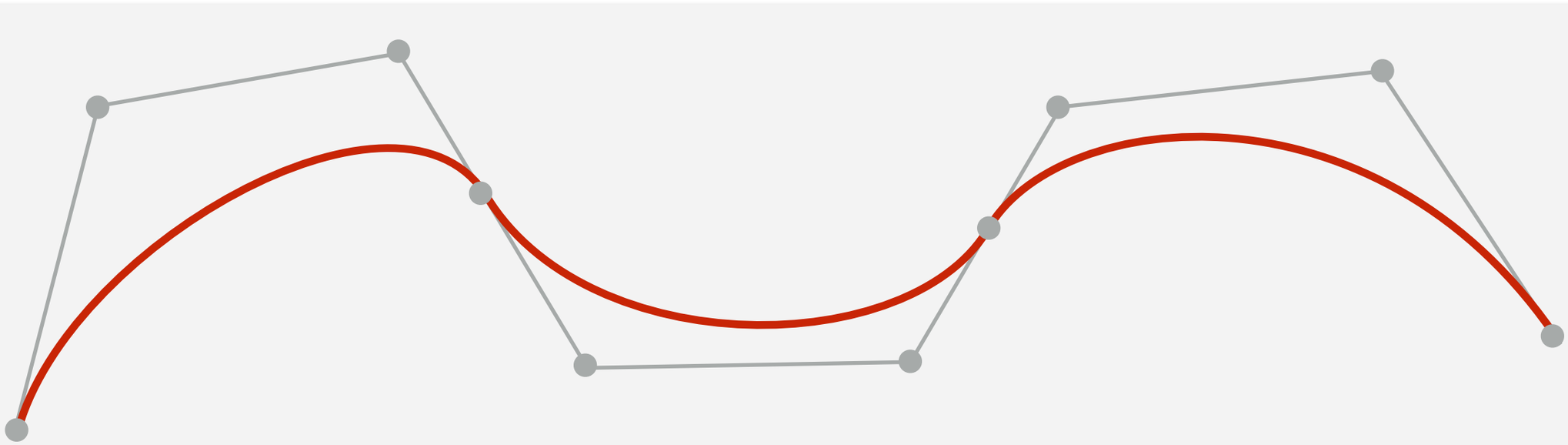
Splines



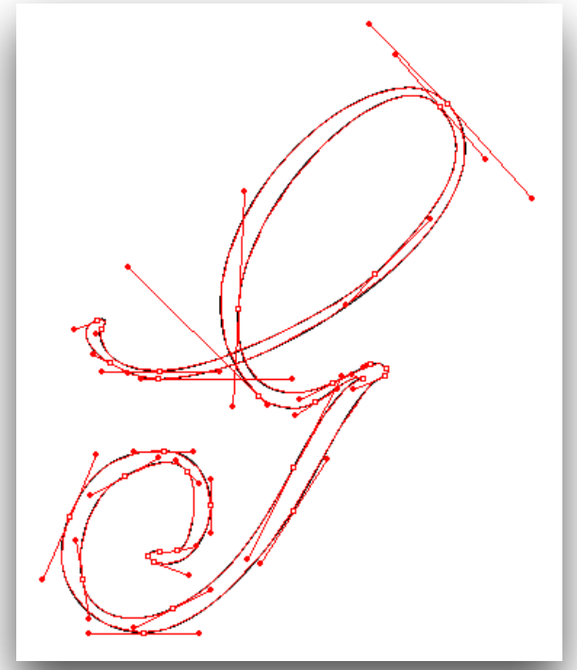
Interpolating - pass through the control points

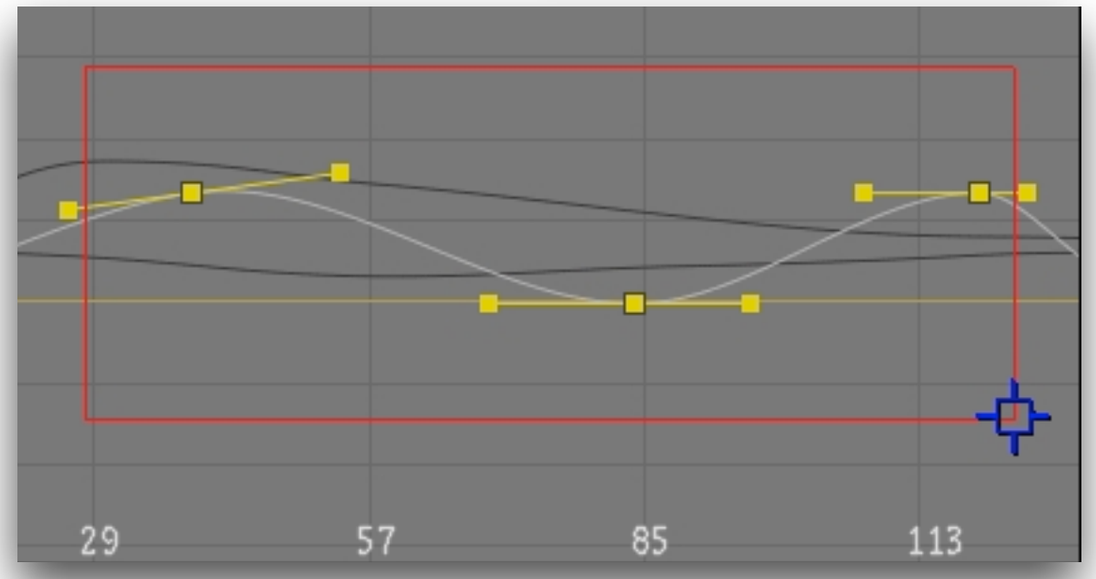


Approximating - guided by the control points

Splines

- **Spline** - piecewise parametric polynomial function
- Join low degree (mostly cubic) splines
- **Advantage:** local control of curve
- Usage:
  - 2D illustration
  - Fonts
  - 3D modeling
  - Animation (keyframes)





Spline specifications

1. Specifying the set of **boundary conditions** that are imposed on the spline

2. Specifying the **matrix** that characterises the spline

3. Specifying the set of **blending functions** that determine how geometric constraints on the curve are combined to calculate positions along the curve path.

Spline specification: Boundary conditions

- Considering a parametric cubic polynomial spline representation:

$$Q(t) = at^3 + bt^2 + ct + d, 0 \leqslant t \leqslant 1$$

- Set the boundary conditions for:
  - endpoint coordinates: Q(0) and Q(1)
  - first derivatives at the endpoints: Q'(0) and Q’(1)
- These four boundary conditions are sufficient to determine the values of the four coefficients a, b, c, and d

Spline specification: Matrix form

- $\mathbf{M_{geom}}$ - four-element column matrix containing the geometric constraint values (boundary conditions)
- $\mathbf{M_{spline}}$ - 4x4 matrix that transforms the geometric constraint values to the polynomial coefficients and provides a characterisation for the spline curve; useful for transforming from one spline representation to another

$$Q(t) = at^3 + bt^2 + ct + d, 0 \leqslant t \leqslant 1$$

$$Q(t) = \begin{bmatrix} t^3 & t^2 & t & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

$$Q(t) = T \cdot M_{spline} \cdot M_{geom}$$

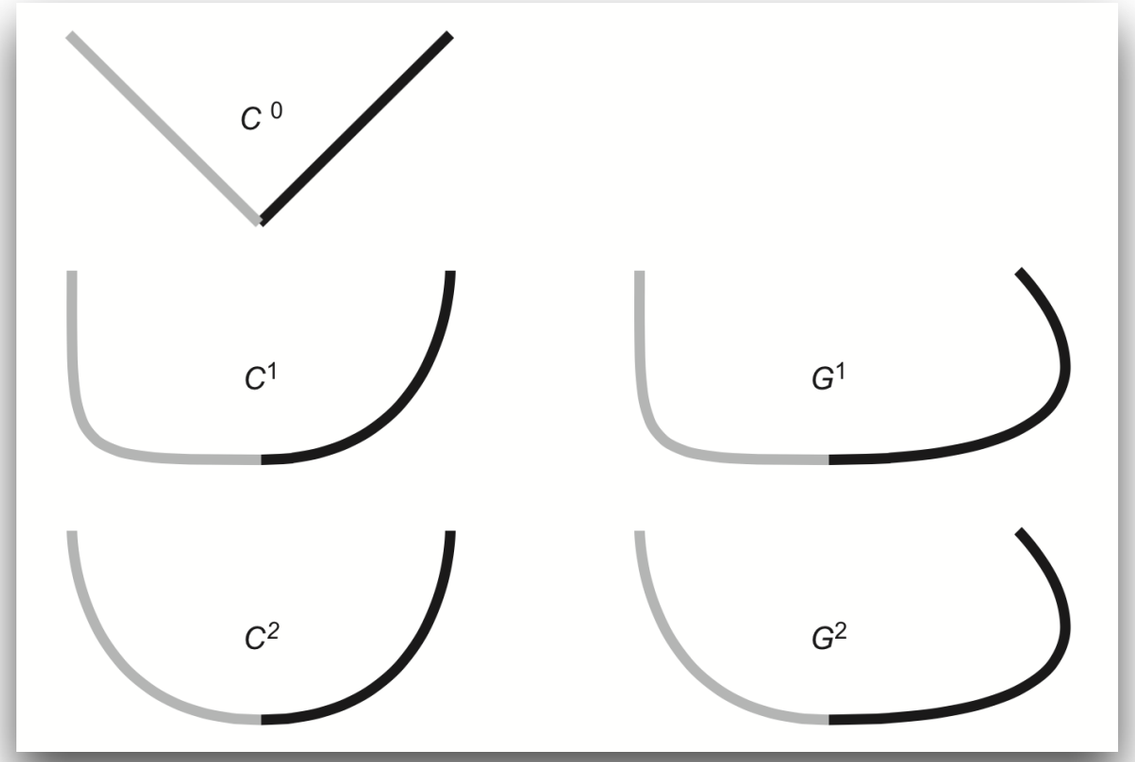
Spline specification: blending functions

- The curve is a weighted sum of the elements of the geometry matrix
- The weights are cubic polynomials of t, called **blending functions**

$$B = T \cdot M_{spline}$$

Continuity at joints

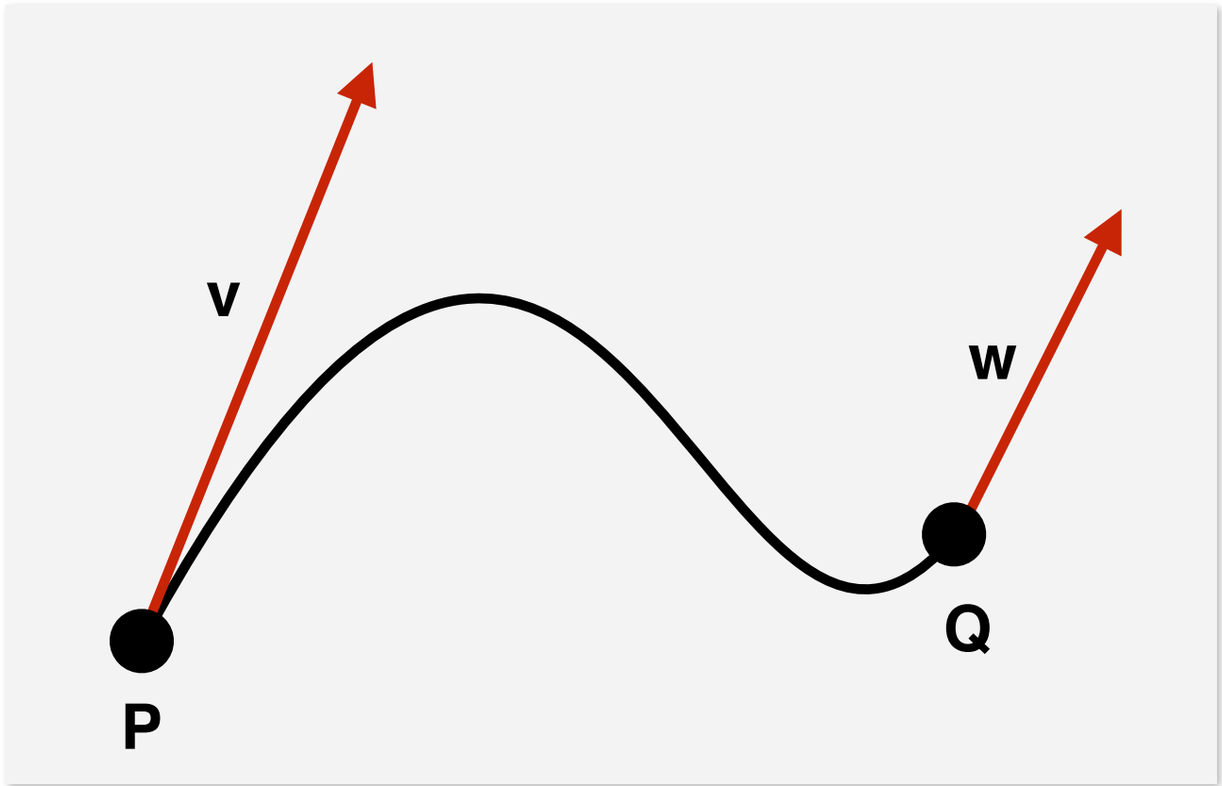
- **C^0 continuity** - spline segments are connected at joints
- **C^1 continuity** - spline segments share same first derivative at joints
- **C^n continuity** - spline segments share same n^{th} derivative at joints
- **G^0** - Same as C^0
- **G^1** - Weaker than C^1, the tangents lengths can be different



[Peter Shirley et al. Fundamentals of Computer Graphics]

Hermite spline

- Named after the French mathematician Charles Hermite
- Interpolating piecewise cubic polynomial spline
- Specified by the **endpoints** and the **tangent vectors** at endpoints (first derivative)



Hermite curve

$$Q(t) = \begin{bmatrix} t^3 & t^2 & t & 1 \end{bmatrix} \cdot M_{spline} \cdot M_{geom}$$

$$\begin{aligned} Q(0) = P &= \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \cdot M_H \cdot G_H \\ Q(1) = Q &= \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \cdot M_H \cdot G_H \\ Q'(0) = \mathbf{v} &= \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \cdot M_H \cdot G_H \\ Q'(1) = \mathbf{w} &= \begin{bmatrix} 3 & 2 & 1 & 0 \end{bmatrix} \cdot M_H \cdot G_H \end{aligned}$$

$$\begin{bmatrix} P \\ Q \\ v \\ w \end{bmatrix} = G_H = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \cdot M_H \cdot G_H$$

Hermite curve

- Hermite geometry vector/matrix

$$G_H = \begin{bmatrix} P \\ Q \\ v \\ w \end{bmatrix}$$

- Hermite basis matrix

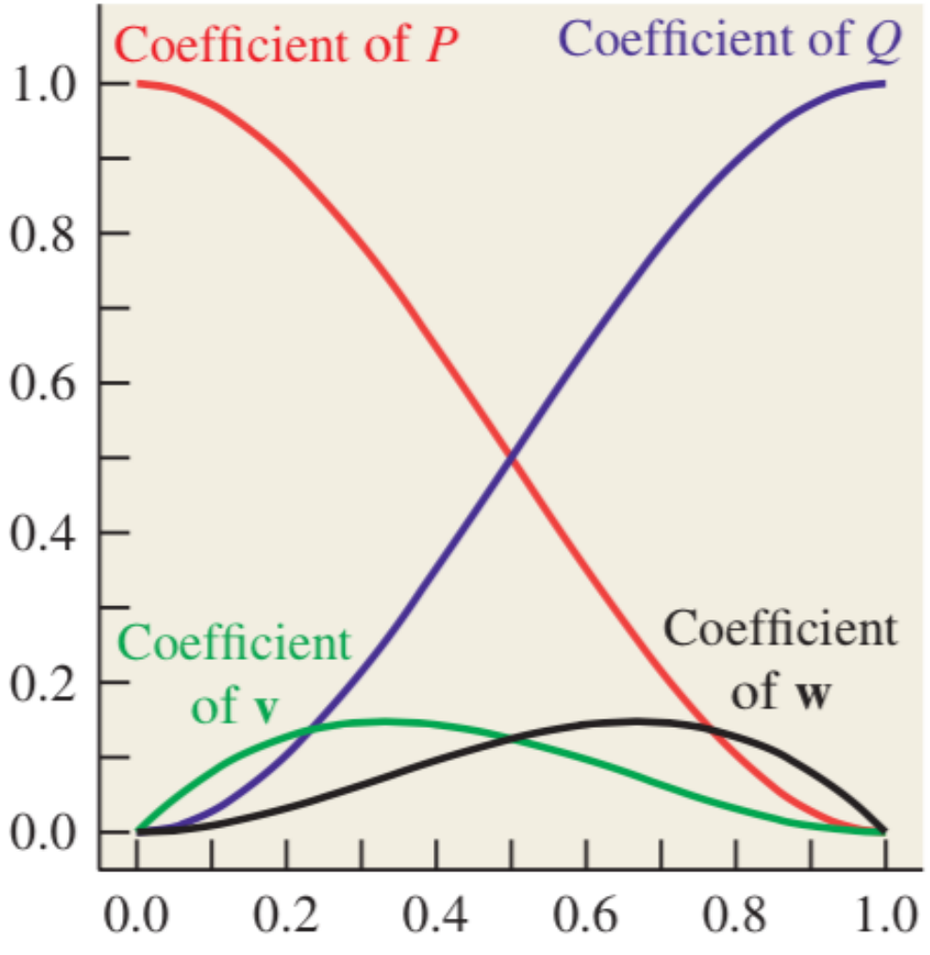
$$M_H = \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Hermite spline

- The four polynomials are called **Hermite basis (blending) functions** and tell how the inputs are combined to make the curve

.

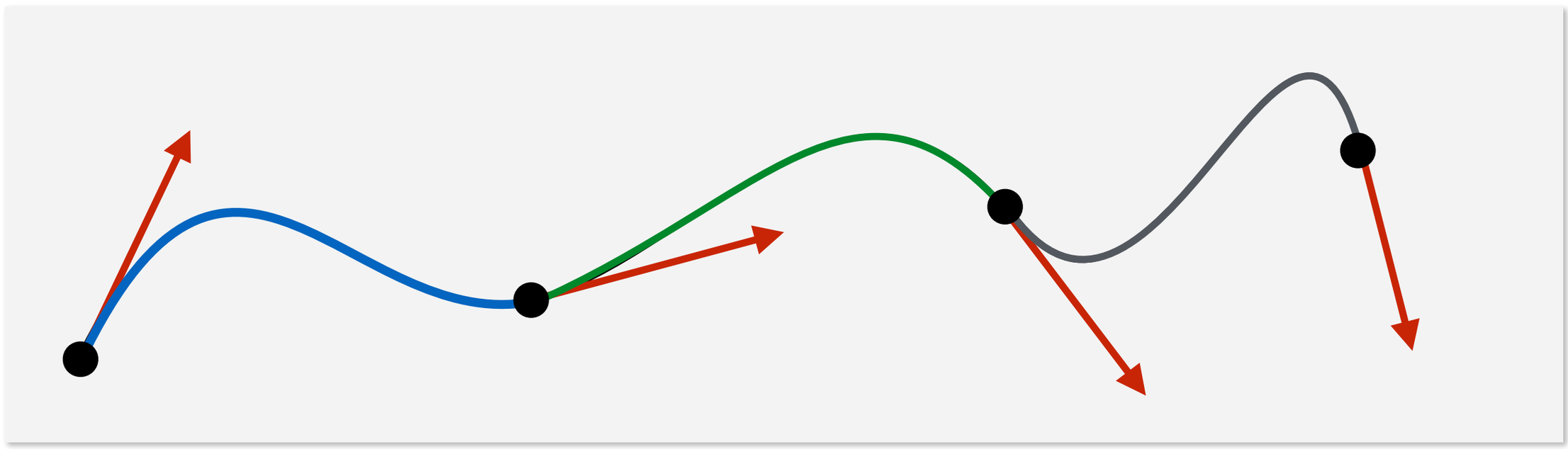
$$f(t) = (2t^3 - 3t^2 + 1)P + (-2t^3 + 3t^2)Q + (t^3 - 2t^2 + t)\mathbf{v} + (t^3 - t^2)\mathbf{w}$$



[John F. Hughes, Computer Graphics: Principles and Practice (3rd Edition)]

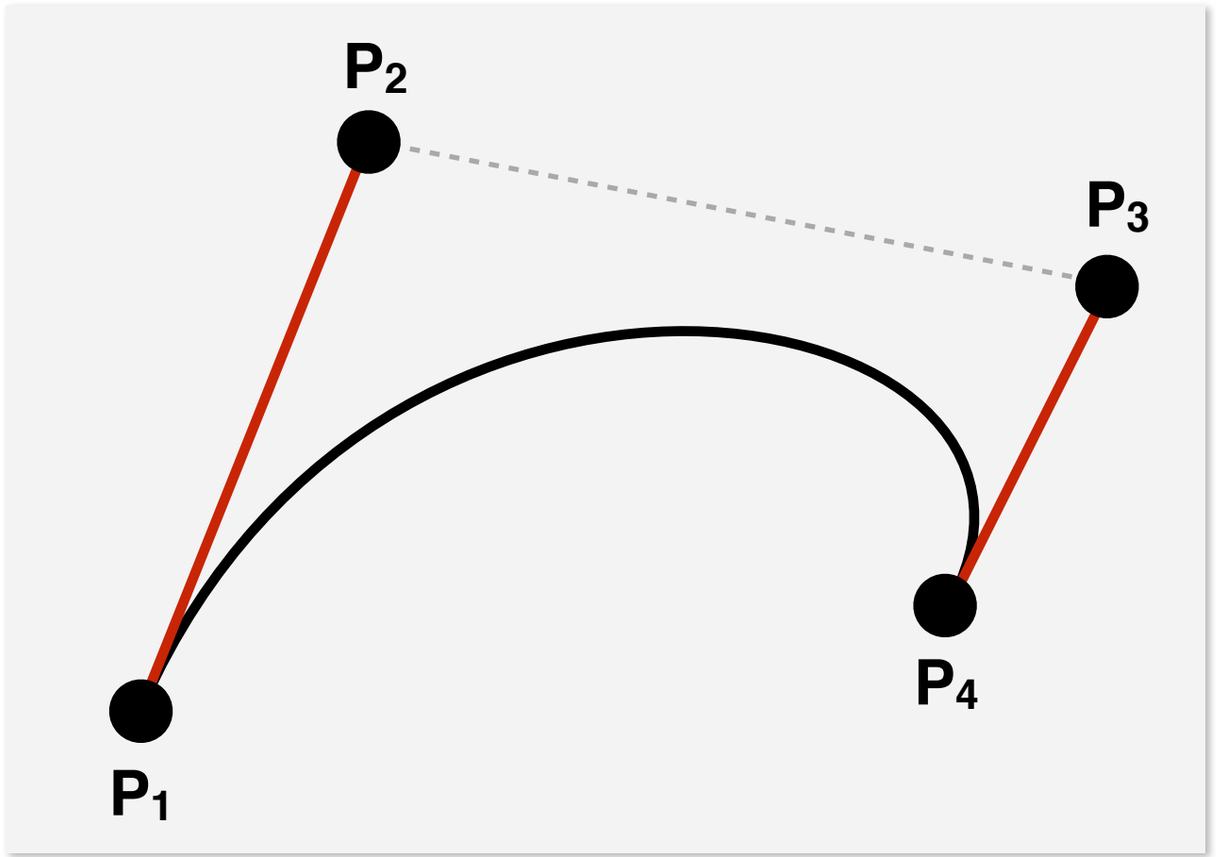
Hermite spline

- Link a chain of curve segments into a C^1 spline:
  - use for the end of the segment and the beginning of the next one the same values for the position and derivative
- For a set of *n* control points a cubic Hermite spline contains *(n–2)/2* cubic segments
- Advantages:
  - **local control** over the shape
  - C^1 **continuity**



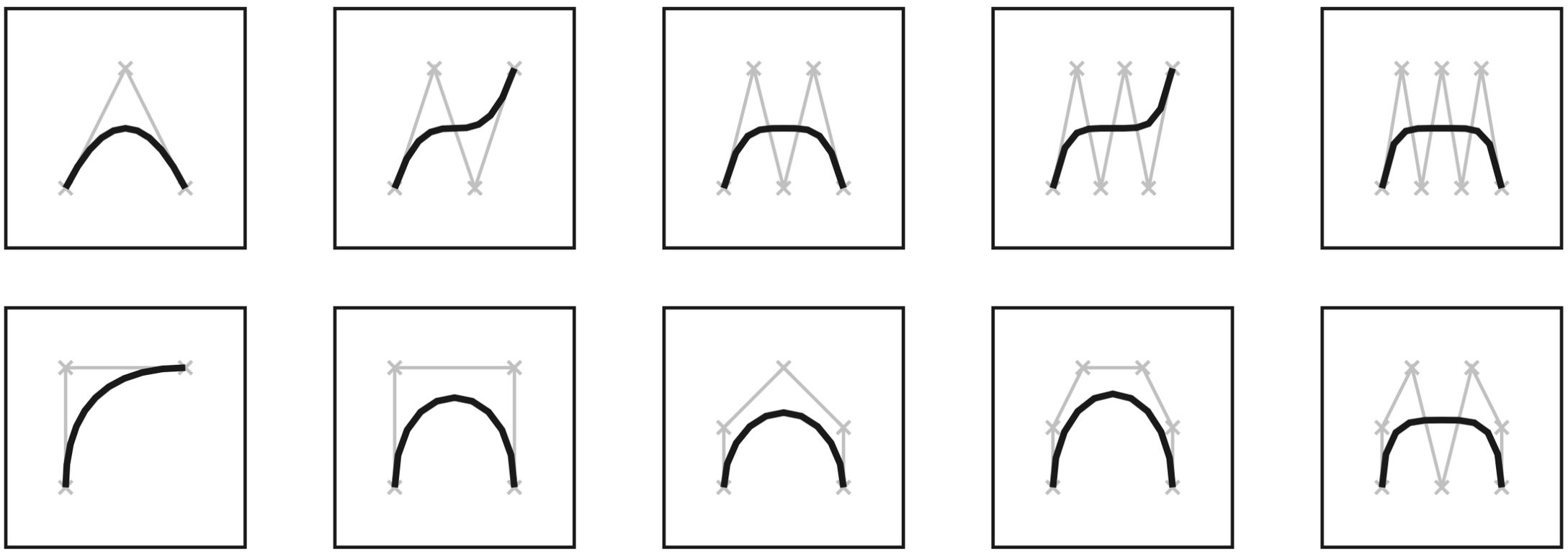
Bézier curve

- Firstly developed by **Paul de Casteljau** at Citroën
- Named after **Pierre Bézier** (from Renault)
- Widely used in:
  - CAD
  - font definition
  - animation
  - graphic design



Bézier curve

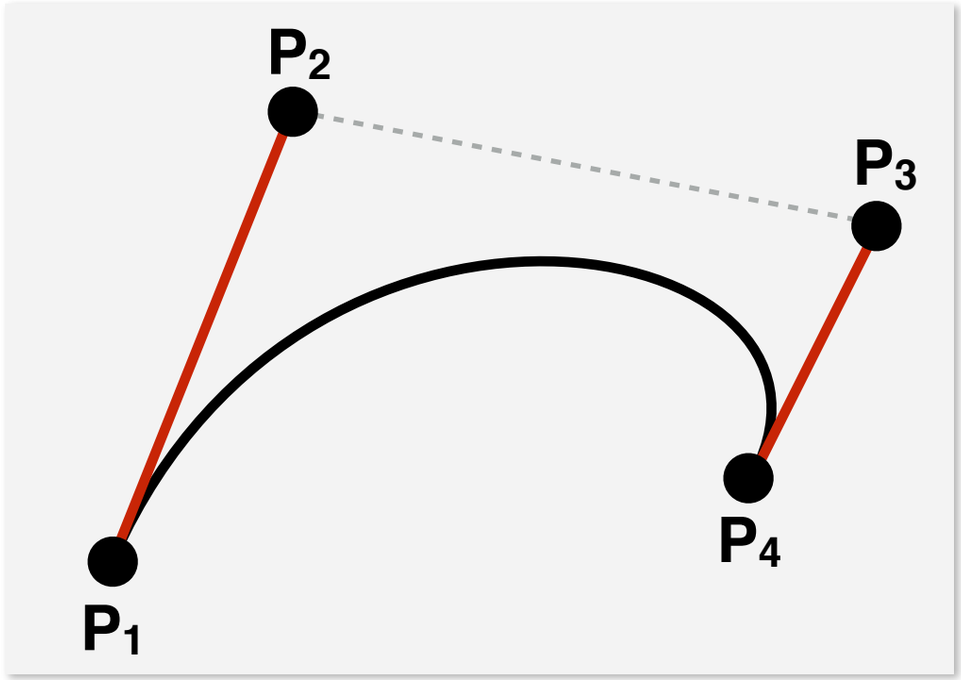
- A curve of degree d is controlled by *d + 1* control points and approximates its control points
- The curve interpolates its first and last control points and the shape is influenced by the other control points.
- Properties:
  - The curve is bounded by the convex hull of the control points
  - The curves are symmetric, by reversing the order of the control points we get the same curve



[Peter Shirley et al. Fundamentals of Computer Graphics]

Bézier curve

- **Cubic** Bézier curves - can define a curve by specifying 2 endpoints and 2 additional control points
- The curve starts at P_1, finishes at P_4, and first derivative at the beginning of the curve is $3(P_2 - P_1)$ and at the end of the curve is $3(P_4 - P_3)$



Bézier curve

$$f(t) = (-t^3 + 3t^2 - 3t + 1)P_1 + (3t^3 - 6t^2 + 3t)P_2 + (-3t^3 + 3t^2)P_3 + t^3 P_4$$

$$M_B = \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

- The blending functions for Bézier curves have a special form that works for all degrees - known as *Bernstein basis polynomials*

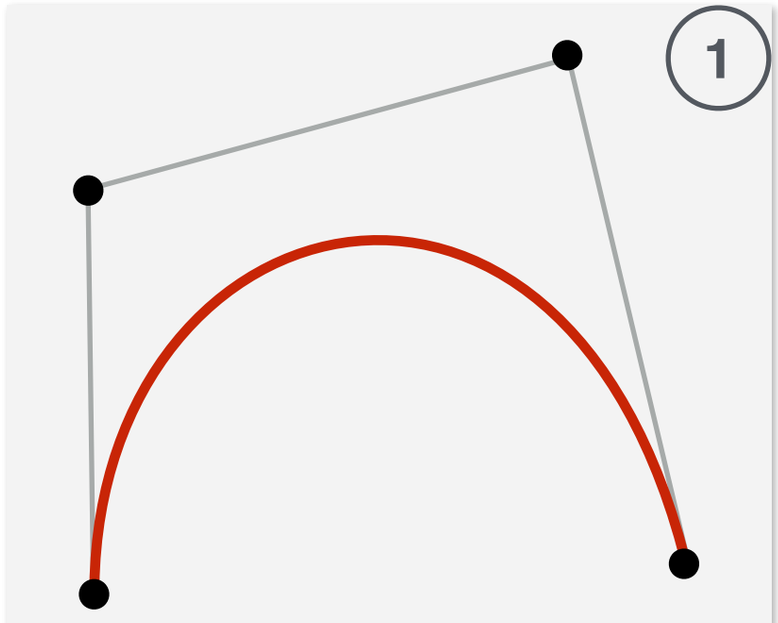
$$b_{k,n}(t) = C(n,k)t^k(1-t)^{(n-k)}$$

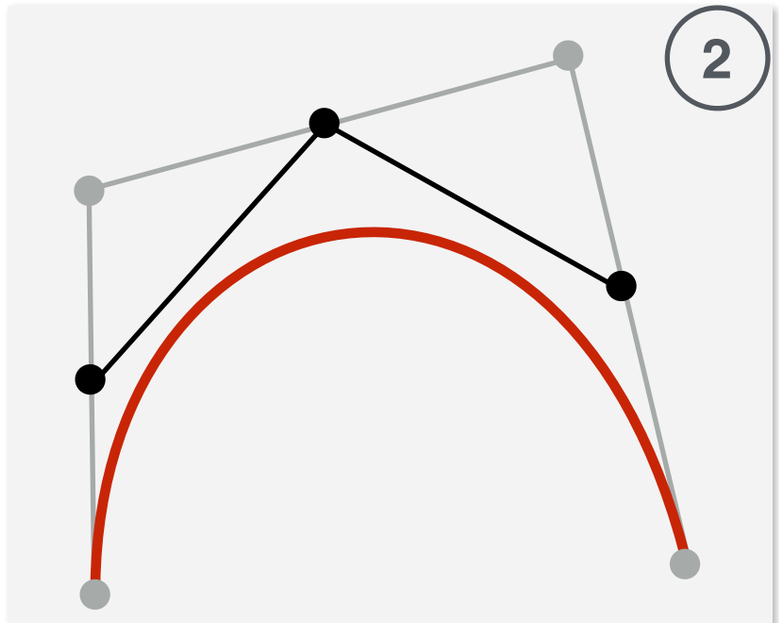
$$C(n,k) = \frac{n!}{k!(n-k)!}$$

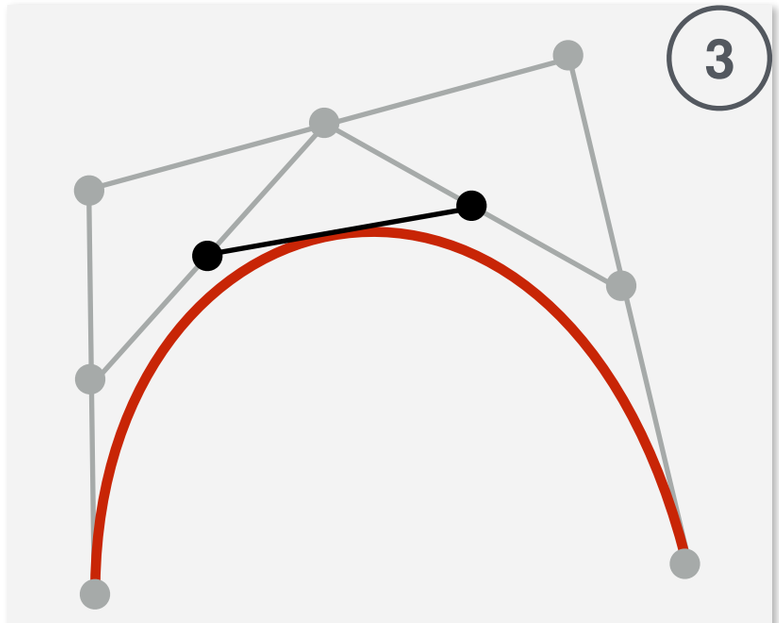
De Casteljau Algorithm

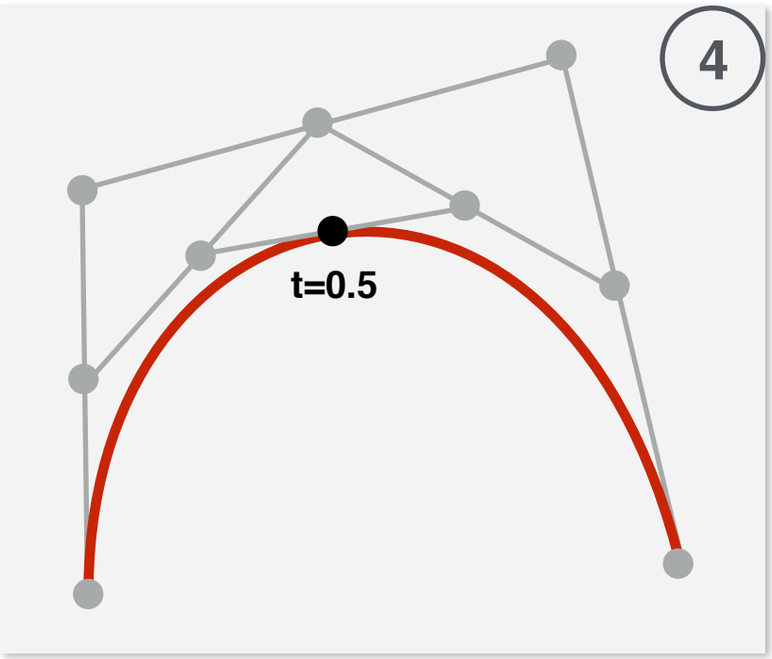
- Uses a sequence of linear interpolations to compute the positions along the Bézier curve of arbitrary order.
- The intermediate points computed during the de Casteljau algorithm form the new control points of the new, smaller segments

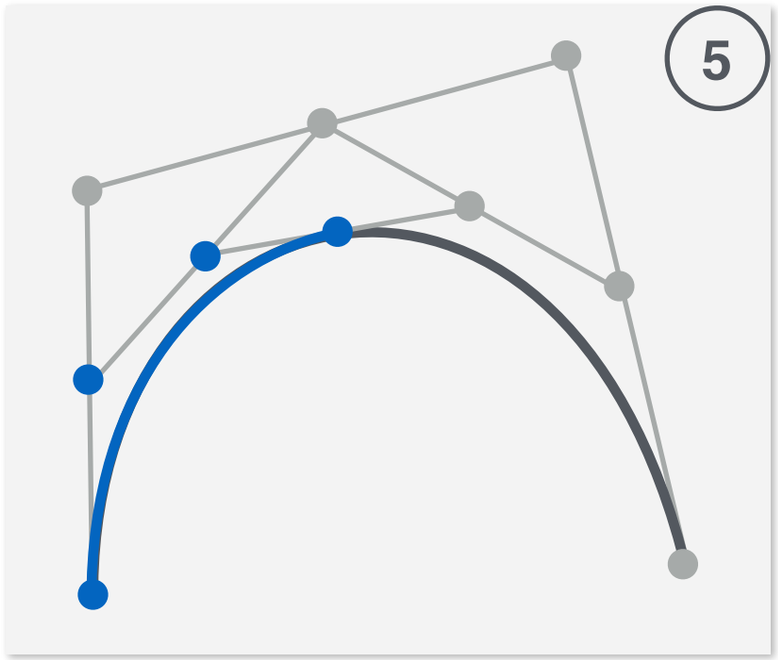
De Casteljau Algorithm

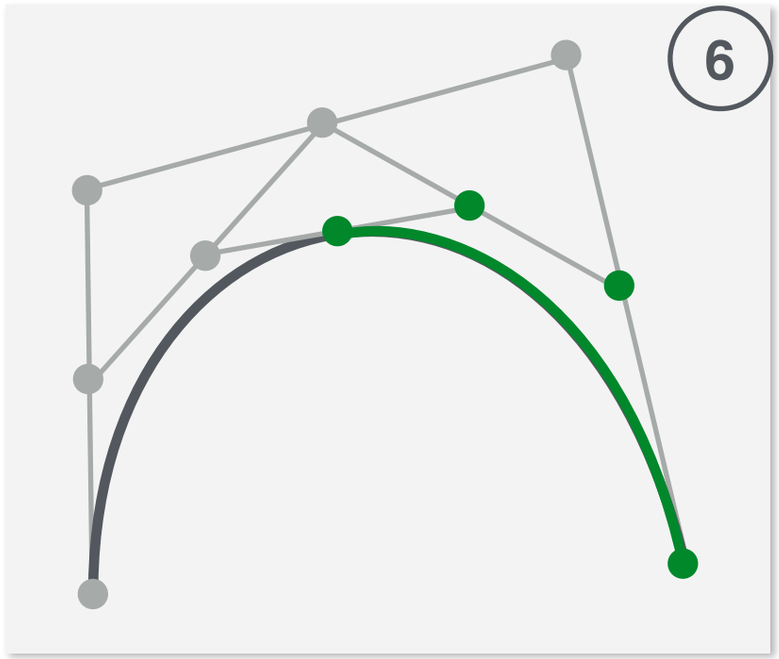








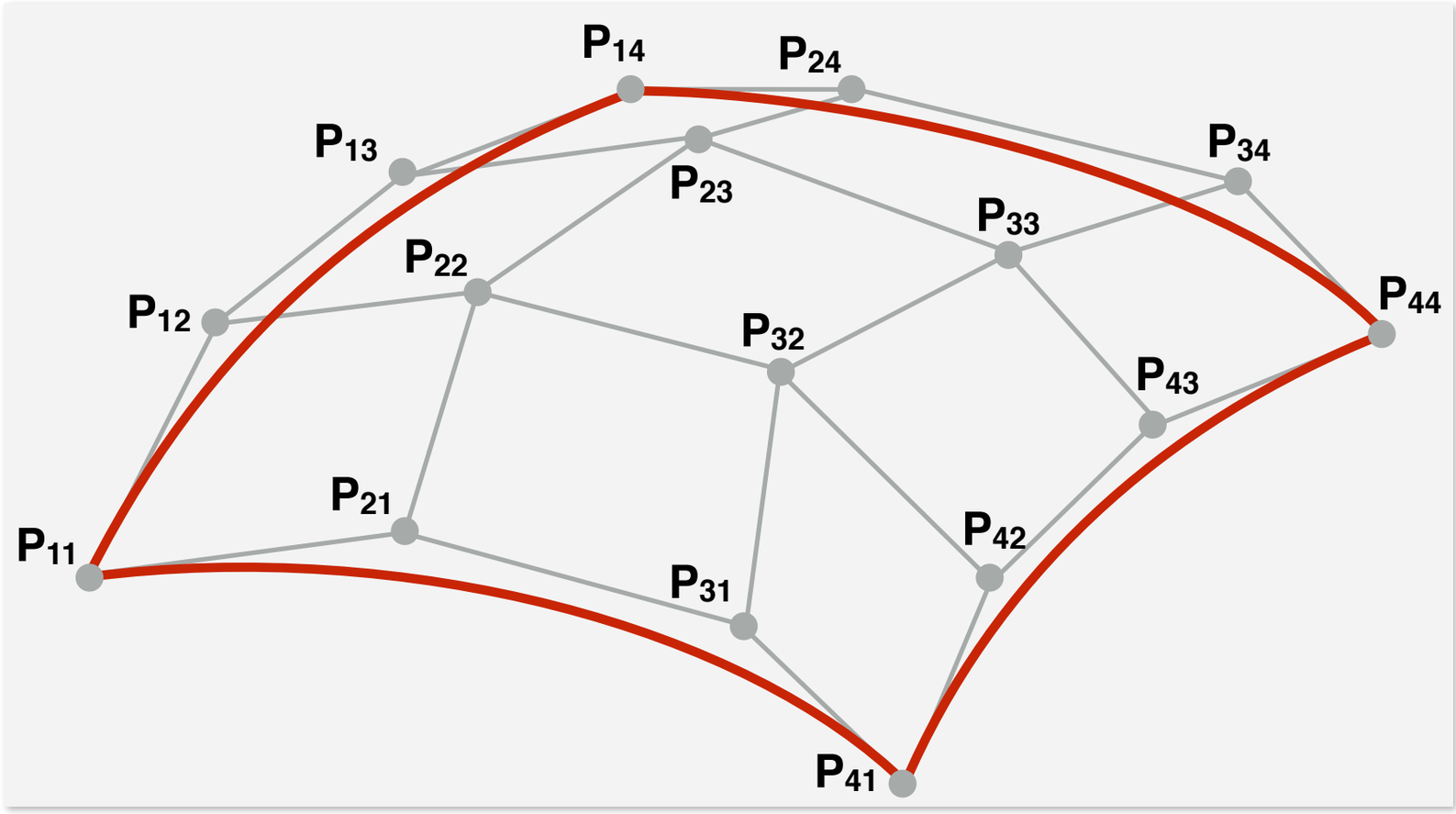




De Casteljau Algorithm

```
point bezierPoint(int n, point[] controlPt, float t)
{
    point deCasPt[n+1];
    for (i=0; i <= n; i++)
        deCasPt[i] = controlPt[i];
    for (r=1; r <= n; r++)  {
        for (i=0; i <= n-r; i++)  {
            deCasPt[i] = (1-t)*deCasPt[i] + t*deCasPt[i+1];
        }
    }
    return deCasPt[0];
}
```

Bézier patches

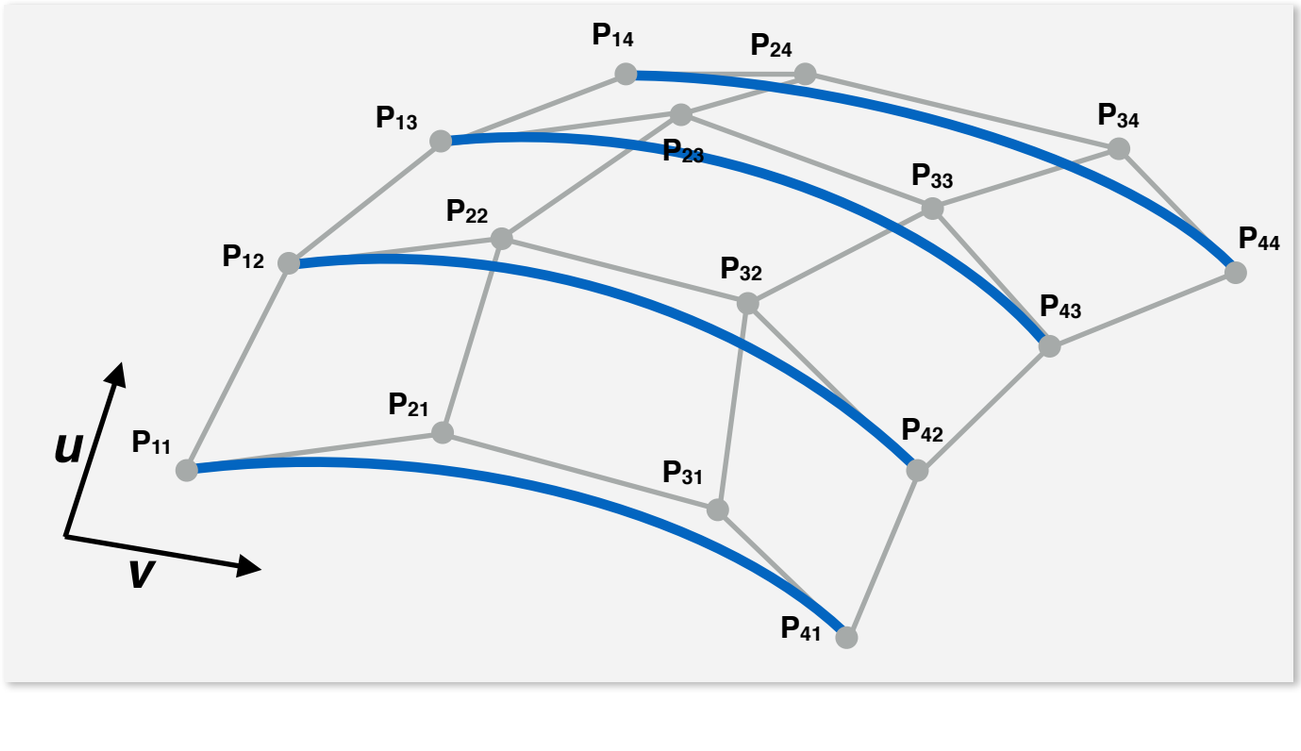


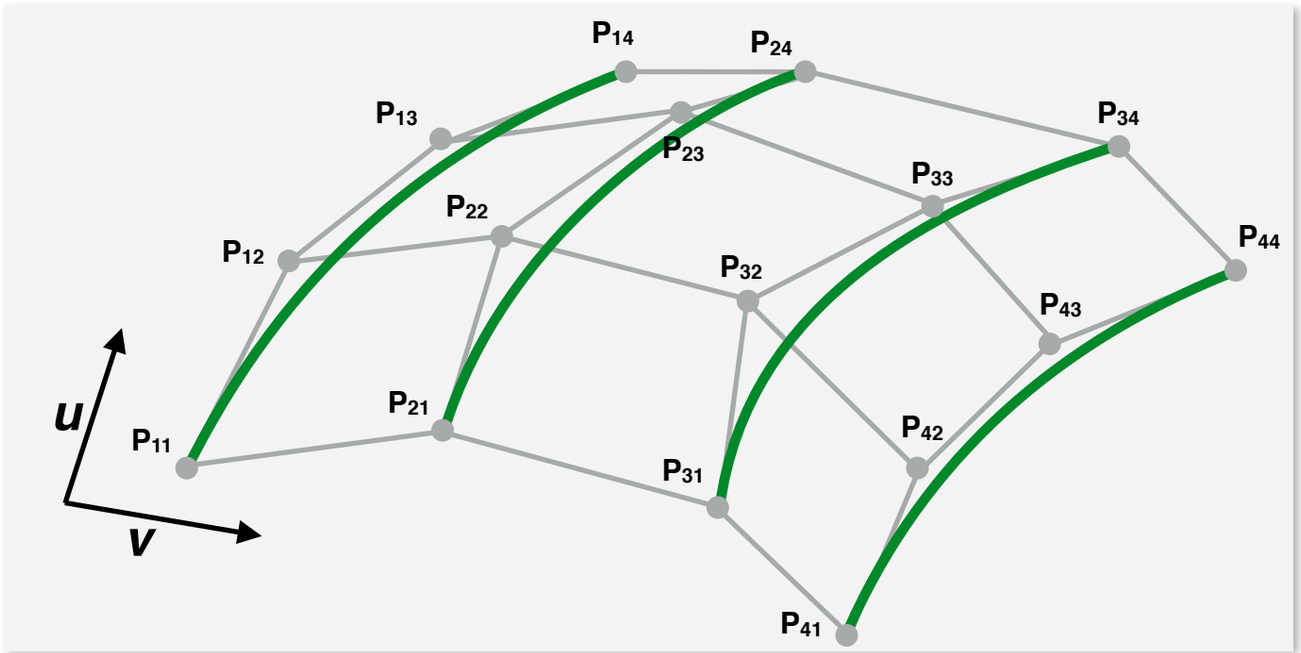
$$Q(u,v) = U \cdot M_B \cdot G_B \cdot M_B^T \cdot V^T$$

$$Q(u,v) = U \cdot M_B \cdot \begin{bmatrix} P_{11} & P_{12} & P_{13} & P_{14} \\ P_{21} & P_{22} & P_{23} & P_{24} \\ P_{31} & P_{32} & P_{33} & P_{34} \\ P_{41} & P_{42} & P_{43} & P_{44} \end{bmatrix} \cdot M_B^T \cdot V^T$$

Bézier patches

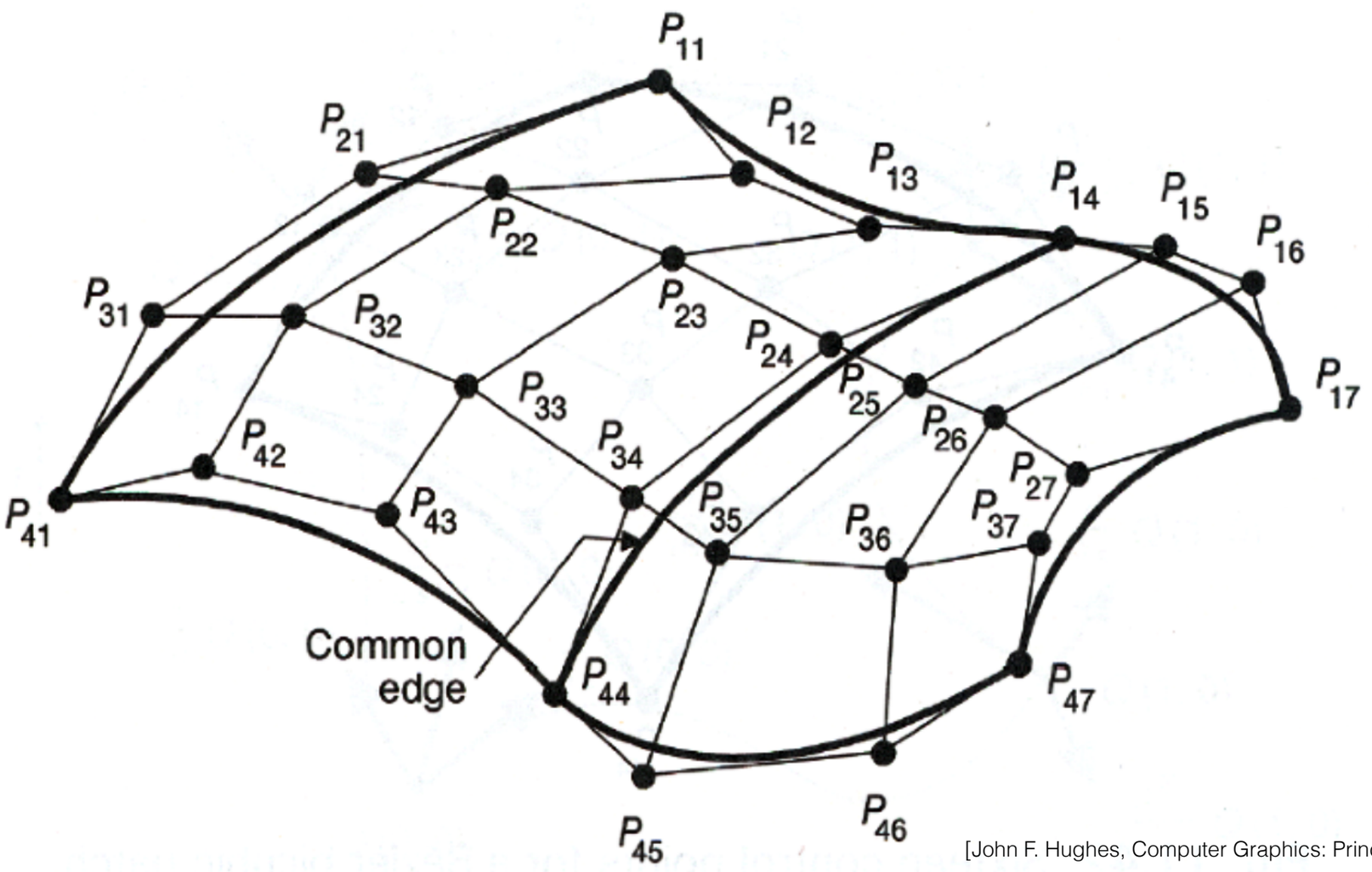
- Plot each curve of constant *u* by varying *v*
- Plot each curve of constant *v* by varying *u*
- *u* and *v* are varying over the interval from 0 to 1





Join Bézier patches

- C^0 - obtained by matching control points at the boundary
- C^1 - obtained by choosing control points along a straight line across the boundary and by maintaining a constant ratio of collinear line segments for each set of specified control points across section boundaries



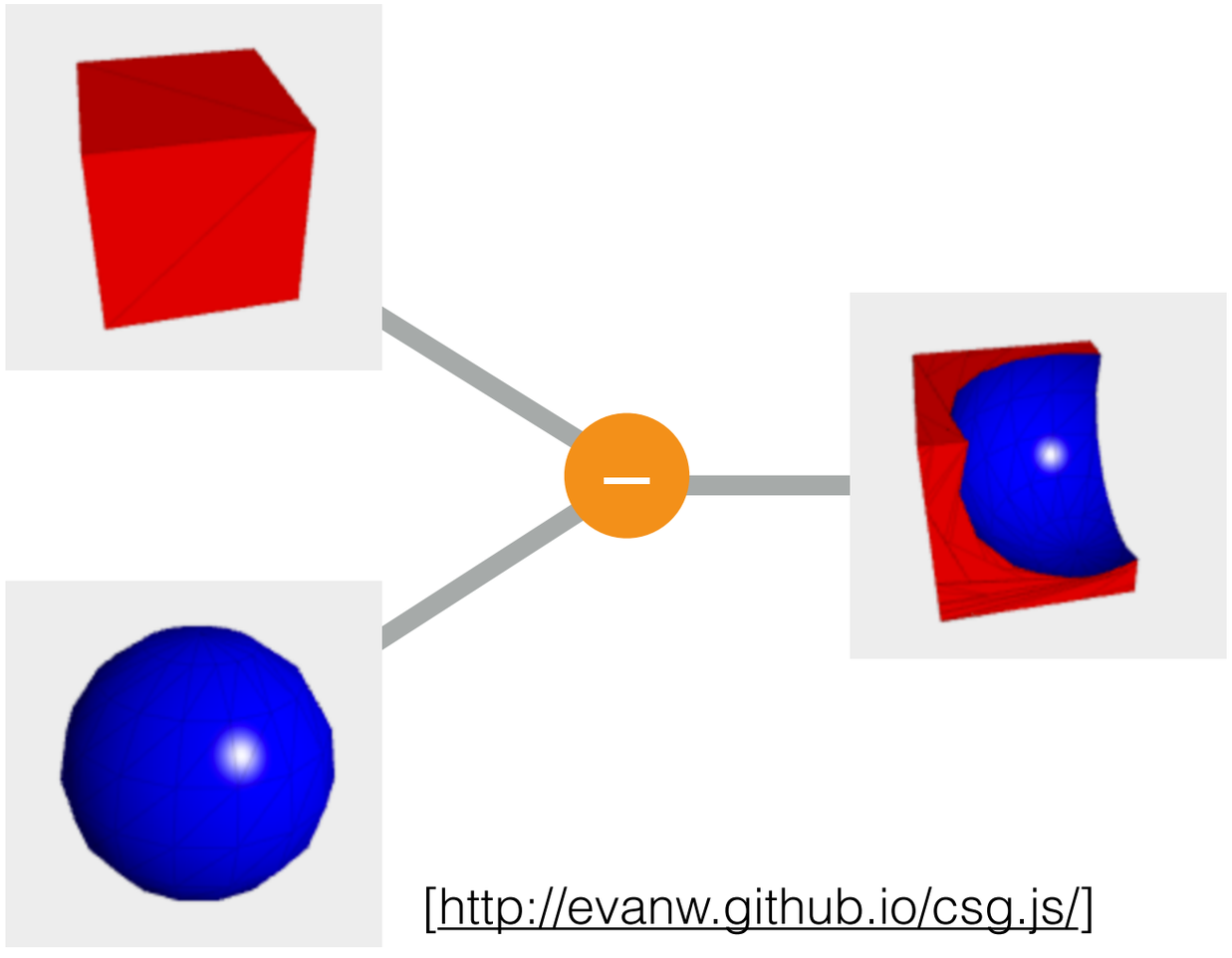
[John F. Hughes, Computer Graphics: Principles and Practice]

3D object representations

Raw data	Surfaces	Solids	High-level structures
Point cloud	**Mesh**	**Voxel**	**Scene graph**
	Subdivision	**CSG**	**Skeleton**
	Parametric	**Sweep**	
	Implicit		

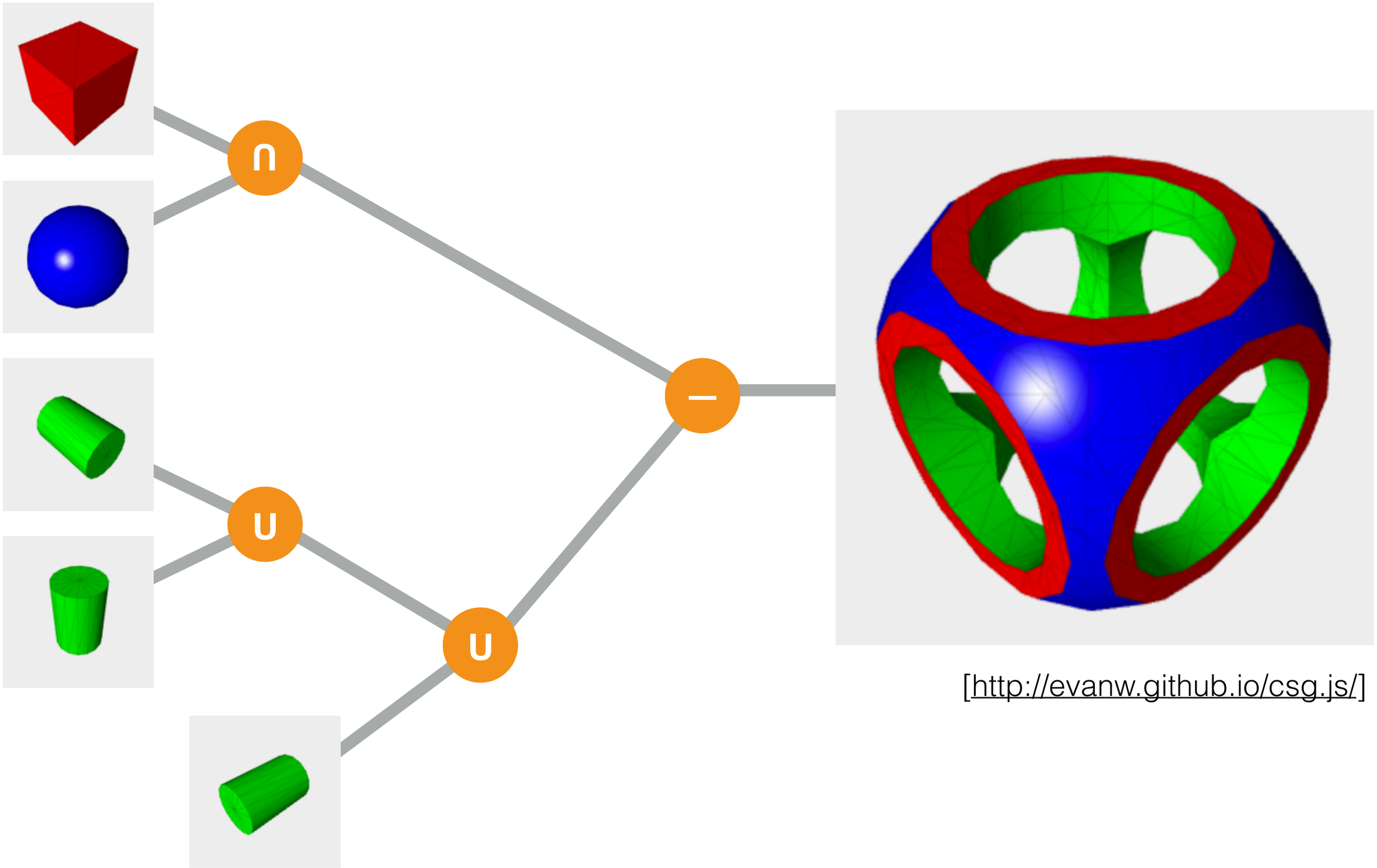
Constructive Solid Geometry (CSG)

- Create complex 3D objects using boolean operators on simple 3D objects
- Represent CSG objects using binary trees
- Operations
  - Union
  - Difference
  - Intersection



[http://evanw.github.io/csg.js/]

Constructive Solid Geometry (CSG)



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